

**Evaluation of Techno-Economic Design and Implementation of
Solar-Wind Hybrid Microgrid System for National University
of Sciences and Technology (NUST), Islamabad**



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(August 2024)

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Supervisor: Dr. Muhammad Yousif

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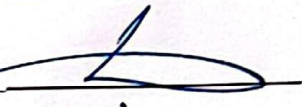
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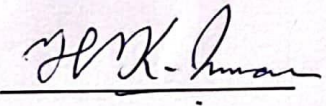
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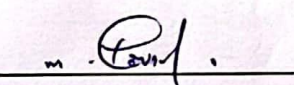
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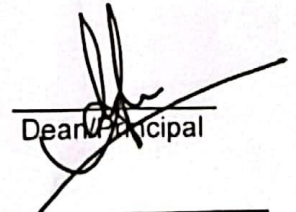
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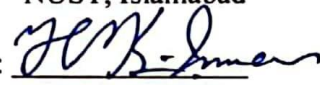
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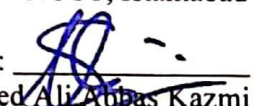
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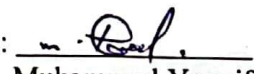
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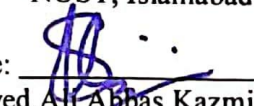
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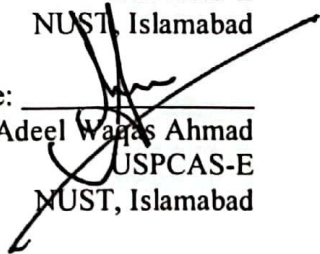
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I dedicate this thesis titled “Evaluation of Techno-Economic Design and Implementation of Solar-Wind Hybrid Microgrid System for National University of Sciences and Technology (NUST), Islamabad” to the professors, supervisor, senior colleagues, and administration of the U.S.-Pakistan Center for Advanced Studies in Energy, whose support and guidance have been contributory in completing this Master's research manuscript.

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LIST OF SYMBOLS, ABBREVIATIONS AND ACRONYMS

RES	Renewable Energy Sources
PV	Photovoltaic
CE	Conventional Energy
MHS	Microgrid Hybrid Systems
CO	Carbon Monoxide
PM	Particulate Matter
CO ₂	Carbon Dioxide
SE	Stirling Energy
RETScreen	Renewable-energy and Energy-efficiency Technology Screening
HOMER	Hybrid Optimization of Multiple Energy Resources
NREL	National Renewable Energy Laboratory
AC	Alternating Current
DC	Direct Current
BESS	Battery Storage or Battery Energy Storage Systems
MPPT	Maximum Power Point Tracker

ABSTRACT

The energy demand and consumption requirements in Pakistan are facing significant challenges due to the recent gradual increase in population; intensifies the critical energy issues confronting the country. Additionally, energy consumption per capita in Pakistan has been steadily rising and is expected to continue increasing in the future. However, the carbon dioxide gas and other toxic gasses from this conventional energy is the main concern for energy generation in Pakistan, South-Southeast Asia and the rest of the world. Hence, Renewable Energy Sources (RES) such as solar photovoltaic (PV) and wind turbines have been contributing a significant percentage of their daily energy needs. However, this energy has not had such standard rather than conventional energy such as fossil fuels regularly. Thus, a microgrid hybrid system containing Solar-Wind renewable and conventional generator needs to produce sustainable Electricity in National University of Sciences and Technology (NUST), Islamabad. On one hand, solar-wind combination can produce energy with specified timing. On the other hand, there are expensive costs. This factor can have prominent issues such as frequent electricity blackouts and imbalanced energy provision. Therefore, the microgrid hybrid systems can evaluate these incorporating solar PV and wind power renewable energy sources and fossil fuel generators as a balance and having an optimum feasibility cost. In addition, this optimum feasibility can provide predictable forecasts of these energy production in the future term based on economic factors in this University, Islamabad. This assessment can have Hybrid Optimization of Multiple Energy Resources (HOMER) software simulation which faced by challenges to explore sustainable solutions through the integration of renewable/conventional energy sources into the existing energy infrastructure.

Keywords: Energy Needs, Microgrid Hybrid Feasibility, HOMER Software Solutions

CHAPTER 1:INTRODUCTION

The power and sustainability of energy production in emerging countries play a dual role, meeting immediate energy demands while contributing to long-term energy development. However, as populations grow rapidly, so does the demand for energy, intensifying both factors. This surge in demand places extreme strain on vital indigenous resources and expensive local goods, leaving imported electricity as the sole survival option [1]. Consequently, the energy desires in developing nations are critically essential for furthering economic growth and enhancing their local energy reserves. Fossil fuels, a conventional energy remain the primary means of electricity generation, despite their significant production costs and associated environmental problems, such as greenhouse gas emissions and climate change [2]. The environmental impact of conventional energy production is highlighted by the emission of greenhouse gases such as Carbon Monoxide (CO) Particulate Matter (PM) and Carbon Dioxide (CO₂) as depicted in Figure 1.1, which illustrates emissions from natural gases, biomass, and electricity worldwide, contributing to ozone layer depletion.

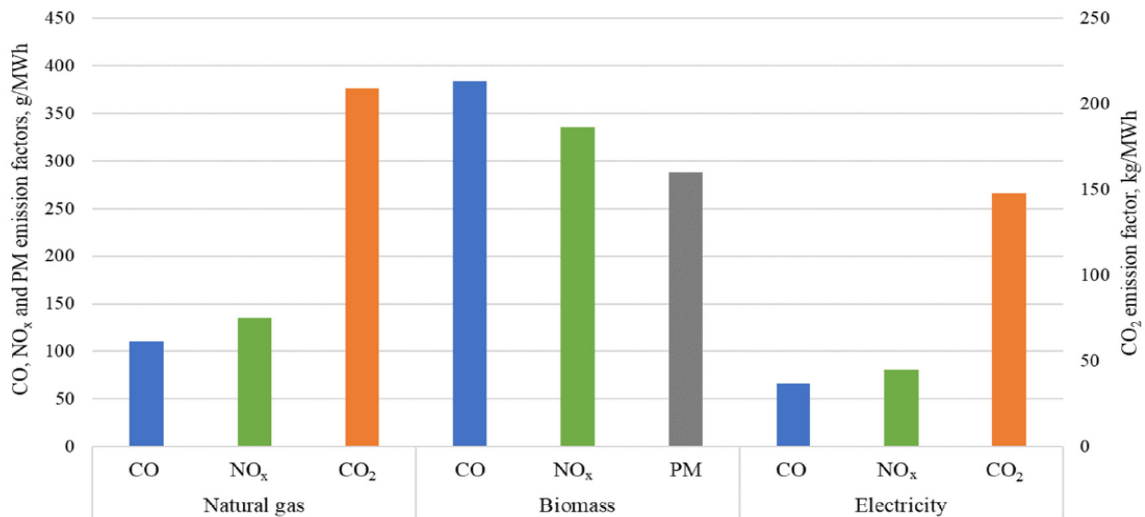


Figure 1.1: Emission Gas Factor

The peak emissions of Carbon Dioxide (CO₂) gases, as depicted in Figure 1.1 (above), are primarily attributed to natural gas and electricity production, contributing significantly to climate change. Additionally, emissions of Carbon Monoxide (CO) gas from equipment production aggravate these climate change effects. The efficiency of energy production, determined by variables such as Stirling Energy (SE) technology, plays a crucial role. SE, an external combustion engine, facilitates the transformation of thermal energy into kinetic energy, further influencing emissions. As the result, these analyses of the gasses are based on emission parameters in Figure 1.1 which reveals so many emissions to the environment ozone layers from biomass and natural gas combustion, as well as power generation. The authors also account for unexpected additional expenses, allocating 20% for natural gas technology and 40% for SE, as determined by feasibility studies [3].

For example, recent reports by Al Jazeera News have connected heavy rainfall in the United Arab Emirates (UAE) and the Sultanate of Oman to fossil fuel emissions. These emissions contribute to climate change, resulting in substantial rainfall precipitation over these countries. The significant emissions of fossil fuels, mainly produced from petroleum used in conventional vehicles, worsen global warming and hence leading to continuous rainfall. The continuous production of crude oil in the UAE and Oman further amplifies heat radiation regularly in their environments, exacerbating global warming [4]. Thus, the challenge for these ongoing crude oil productions is not only for the Middle East but for global policymakers: how can we substitute this conventional energy with sources that emit fewer toxic gases in the future? While crude oil is currently a primary energy source, we could reduce greenhouse gas emissions by exploring alternative, risk-free energy solutions.

1.1. Background

The global climate has undergone substantial changes, with further shifts into predicted matter due to global warming. Climate change presents a complex challenge affecting various environmental sectors and necessitates international cooperation, such as initiatives led by the United Nations Climate Change sector. Rising temperatures in numerous regions, attributed to climate change, represent a crisis that has escalated rapidly since the onset of the industrial revolution. The increasing acknowledgment and

consideration of climatic uncertainties in policymaking at both regional and national levels indicate the impracticality of assessing the precise impacts of climate change on individual sectors [5],[6].

Various societies experience climate change differently due to specific pathways influenced primarily by socio-cultural factors. For instance, while sea-level rise represents a climate-related occurrence that could impact millions of individuals residing near sea levels, its actual consequences depend on site-specific socioeconomic circumstances, such as the availability of resources to manage sea-level rise and reliance on fisheries. Therefore, there is widespread agreement within the scientific community regarding the necessity to enhance comprehension of localized climate change impacts and their experiential nature. Indigenous and local knowledge systems offer valuable insights into localized climate change impacts. Additionally, these systems are well-suited to elucidate how climate change affects local socio-ecological systems, livelihoods, and cultures, given their extensive and detailed observations of these systems over the long term [7],[8].

1.2. Conventional Energy Over-Use

The disparities in electricity accessibility between rural and urban areas are significant, with rural regions often facing higher costs. Rapid global increases in electrical energy consumption have depleted conventional energy fossil fuel reserves used for production. Additionally, various energy factors, including insufficient planning, political unrest, and a lack of institutional capacity, contribute to power crises in emerging nations. The increasing global population, particularly in developing nations, along with technological advancements like the mass production of silicon chips, has triggered a staggering surge in energy demand, especially in fossil fuel consumption. This conventional energy source, being finite in nature, carries substantial environmental repercussions, contributing to the escalating phenomena of global warming and climate change [9],[10],[11],[12].

Recently, researchers worldwide have intensified their quest for innovative technological solutions aimed at reducing or eliminating reliance on fossil fuels. One

promising approach involves enhancing the efficiency of conventional power conversion mechanisms through the implementation of waste heat recovery systems. This strategy not only addresses fossil fuel consumption but also mitigates its detrimental impact on climate change [13],[14],[15]. Although waste heat recovery systems offer one solution, there are many other approaches too to reducing the fossil fuel consumption use and mitigating the greenhouse effect. However, challenges remain with the power markets and feasibility economy in this energy.

For instance, a conference symposium organized by the Ministry of Energy (Petroleum Division) in Pakistan, held at U.S-Pakistan Center for Advanced Studies in Energy’s seminar hall, National University of Sciences and Technology (NUST), focused on natural gas policies and infrastructure. Natural gas, a significant component of fossil fuels, is a crucial aspect of the global energy market due to its availability as a conventional energy source. Figure 1.2 illustrates the current and projected gas supply and demand trends in Pakistan. Known for its lightweight and efficient performance in machinery (as exemplified by Qatar's Natural Gas in Ras Laffan Region – north of Doha passing Al Khor City), natural gas emits fewer greenhouse gases compared to other fossil fuels. While it serves as a sufficient energy source, its consumption also poses challenges related to greenhouse gas emissions, which could impair environmental issues in the future [16]

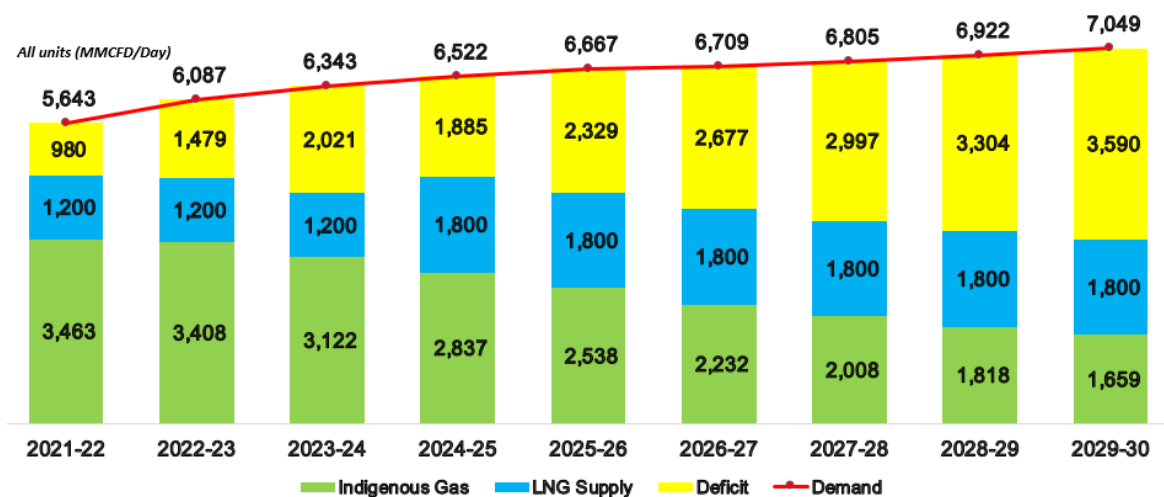


Figure 1.2: Supply and Demand of Pakistan Gas' Facilities (Source: Symposium, Pakistan’s Ministry of Energy)

Based on the data presented in Figure 1.2 (above), Indigenous gas reserves in Pakistan are projected to decline significantly over the next decade. This indicates a heavy reliance on imported gas from other countries, as domestic reserves are insufficient for future use. However, there appears to be limited interest from local buyers in imported gas due to inadequate local goods for sale in energy markets. This instability in the energy markets has led to an increase in demand for gas only among local buyers, as highlighted by the continuous red markings in Figure 1.2. Consequently, the gas deficit in Pakistan is expected to gradually increase over the next 10 years. Although natural gas could be a solution for Pakistan, all its facilities currently depend heavily on conventional energy sources, sidelining the benefits of renewable energy. This statement underscores the critical need for improving energy stability through renewable energy feasibility in Pakistan.

1.3. Electricity Energy Reluctant

An essential energy challenge in numerous regions, including remote rural communities, stems from limited access to the electricity sector. Despite recognition of electricity access as a fundamental need, isolated rural areas distant from the central national grid face practical challenges in energy provision due to economic constraints. These challenges include higher associated costs, concerns about energy quality, security issues, and energy losses related to extending the grid to remote areas [17]. These issues present substantial obstacles to developing electricity infrastructure in remote regions of Pakistan.

Many developing nations in South and Southeast Asia struggle to balance energy supply and demand due to under-developed generating sectors or inadequate transmission capacity, leading to electricity shortages. These nations often experience frequent power outages due to numerous factors, prompting developed nations to focus on renewable energy and technology to promote sustainable development and environmental conservation.

1.4. Renewable Energy Solution

Consequently, renewable energy emerges as a viable solution for these developing nations, offering affordability, accessibility, cleanliness, and ease of use across various forms. Solar, geothermal, wave-tide, biomass, and wind power turbines have proven to be productive energy providers and alternative sources of electricity. Among these, wind power and solar energy stand out due to their cleanliness and unlimited potential. These two renewable energy sources can reduce reliance on fossil fuels and mitigate energy price fluctuations, not only in Pakistan but globally. This perspective is particularly important as crude oil reserves are projected to diminish in the future, challenging their role as the primary energy provider. Figure 1.3 illustrates the renewable energy factors derived from natural resources, highlighting their potential as sustainable energy solutions.

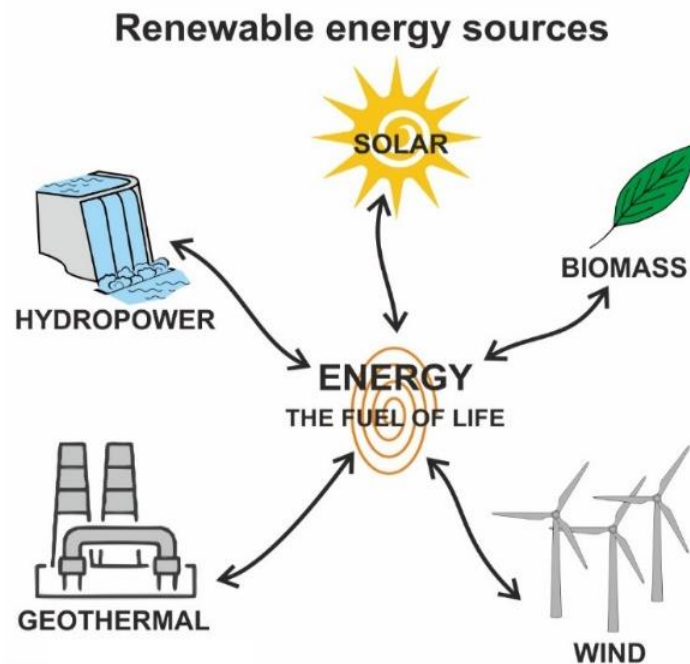


Figure 1.3: Renewable Energy Factors (Source: [Polarpedia](#))

1.5. Microgrid Hybrid Implementation

Decentralized renewable energy systems, such as microgrid hybrid systems, offer a solution to the energy demands of remote communities, potentially improving living standards. These systems promote feasible and efficient energy structures, positively

impacting social dynamics. The affordability and reliability of decentralized renewable energy systems are advantageous for rural areas worldwide, providing sufficient energy without the need for extensive grid extensions. Moreover, they offer a sustainable alternative source to traditional biomass and kerosene liquid currently used in rural regions. Solar PVs and wind power turbines complement each other, balancing benefits and minimizing negative effects. Their hybrid combination addresses availability concerns and provides financial advantages when coupled with storage cell systems. The hybrid microgrid, incorporating both solar and wind power, tackles availability and storage challenges, facilitating with financial gains. Thus, the decentralized renewable energy with hybrid microgrid system could be the next energy provider in the future instead of the primarily fossil fuel source.

CHAPTER 2: LITERATURE REVIEW

2.1 Literature Review

My research thesis explores the literature on Microgrid Hybrid Systems in renewable energy demand and their transformation into the electricity domain. From a wide selection of available articles, I have selected several significant publications yet effectively strong that highlight the applications of renewable energy demand and microgrid hybrid factors across various modern contexts. The economic factors related to energy demand are further analyzed through different sub-topics. For detailed references, readers are encouraged to consult the reference section of my thesis.

This thesis synthesizes elements from power engineering, focusing on Clean Energy Generation, Integration and Storage, and the Smart Grid Architecture, which form the core of my research analysis. In the dominion of microgrid hybrids, scholarly materials specifically investigate renewable energy sources such as Solar Photovoltaic (PV) and Wind Power, which produce energy without a carbon footprint ton weight. Relevant articles also discuss the harmful gases emitted by conventional energy usage, which pose a risk to future generations. The economic aspects, including detailed calculations and future forecasts, aim to balance conventional and renewable energy usage in microgrid hybrid space feature. However, these economic implications of energy usage are a significant concern for stakeholders and businesses promoting these technologies.

My research primarily centers around renewable energy in smart grids, coupled with microgrid hybrid design and techniques. Among the reviewed articles, many focus on methodologies for both conventional and renewable energy only. I am currently identifying options that align with the objectives of my thesis. Several articles have investigated and inquired these questions into the application of microgrid hybrid techniques in the analysis and furthermore interpretation of these factors as well.

2.2 Globally Energy Challenges

Silva et al. describe numerous challenges which faced difficulties in energy globally to accessing electricity demand, particularly in remote rural communities. Despite electricity being recognized as a fundamental right by most governments, these remote rural areas are often far from the central national grid, making energy provision economically unfeasible. These challenges include high electricity costs, concerns about energy quality, and losses incurred in supplying electricity to these remote areas.

Similarity, Ullah et al. highlight the electricity crises faced by developing nations, attributing them to various factors including inadequate power planning and policies. This issue is further entangled in political disputes surrounding power generation in these nations, alongside rapid economic and technological growth. The lack of access to electrical energy impacts a significant portion of the global population, with a disparity between these countries' urban and rural areas.

2.3 Power Quality Disturbances

Nakhai et al. have emphasized the ongoing challenges related to accessing and running of water systems in the United States of America. This country struggles with repairing and maintaining its water systems, leading to over two million Americans experiencing inconsistent access to purifying in clean water. Remote and low-income communities in the USA face even more significant obstacles, as the limited accessibility of the water grid system makes it costly to provide water sanitation to the customers and maintain their livelihoods. Therefore, the solar PV system, functioning with a DC motor regarding DC-DC converter, is strategically engaged to enhance the water purification process. Utilizing DC systems offers the potential to minimize power conversion losses, enhance power quality, and lower power system expenses. These solar PV DC motors are comprehensively implemented for applications requiring speed or position control due to their straightforward speed/torque regulation and exceptional drive capabilities[18]. Consequently, this renewable energy source is associated with addressing the challenges encountered by people in remote regions and providing essential utilities at present.

According to Raza et al., the power sector in developing countries is crucial for meeting current needs and facilitating future progress. As populations grow and urbanization living standard accelerates, the demand for this energy increases, placing significant strain on domestic resources. Consequently, many developing nations rely on imported electricity or costly fossil fuels to meet their energy needs. While these sources may provide a temporary solution to the energy crisis, they often result in additional expenses for consumers and fail to adequately meet demand. Furthermore, under-developed generation infrastructure and transmission capacity issues can lead to power shortages and blackouts in these countries.

2.4 Renewable Energy System

Renewable energy, following extreme situations as described above, emerges as a promising solution for these developing countries because of reasonability, attainability, hygiene, and ease of acquisition in various forms. These renewable energy sources serve as strong contenders, leading to reduced dependence on fossil fuels and crude oil, thus contributing to stability in energy prices. Decentralized renewable energy systems, distinct from conventional grid-based electricity applications, offer feasible solutions in terms of both financial cost and consistency usages. Moreover, renewable technologies such as solar PV monitoring systems and inverters provide better alternatives than the current conventional electricity sources, like kerosene oil and traditional biomass, commonly used in rural areas.

Silva et al. further explains that the decentralized flow of electrical supplies, regards the debate solution in the form of the microgrid systems presents a feasible way for meeting this energy demands of the remote communities. Although this procedure has increased its power system, the power grids while functions can increase their costs. These costs can be also associated with extending forms from the central grid to economically inactive regions in developing nations. As a result, these microgrid systems can offer potential benefits for improving the living standards as well as fiscal costs of these rural populations.

2.5 Microgrid Hybrid System

Wind power and Solar PV, two of renewable energy forms regarding on their hybrid microgrid systems outputs as Raza et al. express that these are a favorable combination nowadays. These forms of renewable energy are both supplement to each other in their advantages. They are also attributing in their attenuate drawbacks, associated with renewable energy exploitation. The hybrid renewable energy as mentioned in the introduction phase comprises wind and solar power consecutively. These wind power and solar PV are considered to combat with availability issues alongside with storage system in electricity and hence for economic benefits.

2.6 Financial Cost Limitation

However, Raza et al. is expresses as constraints regarding the renewable power plants which have a higher construction cost for financial terms. This expensive cost feature further affects its future distribution and maintenance, making it a tough one to sell to stakeholders and investors. As the electricity obtained is of low intensity to comparison to conventional fuel sources, planning for this renewable energy such as wind turbines is a crucial step for enhance its efficiency and electric production. Hence, the cost for the renewable energy output can be such optimize solution of their usage in the rural areas being function of hybrid microgrids as their energy provider.

CHAPTER 3: SIMULTION SETUP

3.1 RETScreen Software Use

The utilization of renewable energy sources and most importantly hybrid microgrid systems regarding solar PV and wind power turbines can be involved with various software tools. These software applications when improvement of computer science fields is a better way for this renewable energy simulate as its calculations and forecasting methods in the future period. One such tool for this statement is the Renewable-energy and Energy-efficiency Technology Screening (RETScreen) software. This software can focus on managing clean and sustainable energy. It is also facilitating low-carbon planning, implementation, emission, monitoring, and reporting.

As highlighted by Paradongan et al., the RETScreen computer software is a machine's proficient duty for assessing the world-wide economic feasibility in energy. This assesses duties can lead to both large-scale and small-scale facilities including solar PV power installations. Equation 3.1 (below), used to evaluate energy generation, integrates a critical variable called the Capacity Factor. This variable is integrated into the software's modeling on the energy factors. By calculating a power plant's output relative to its maximum capacity, the RETScreen software can offer valuable intuitions such as machine learning algorithms based on renewable energy production and gas emissions[19].

$$Capacity\ Factor = \left(\frac{Net\ Generated\ Electricity\ Energy}{Nominal\ Capacity} \right) \times 100 \quad (3.1)$$

The RETScreen Expert software, as defined by Iweh et al., is a computer-based tool, designed to modernize the management of clean energy. This software furthermore is a highly effective analytical resource created by the Canadian government. The basic of this computer-software is for performing techno-economic factors and environmental assessments too in the energy systems. Additionally, RETScreen software is accessible for evaluating photovoltaic (PV) plants which is the main research analysis, allowing for the

comparisons between the government power tariffs and various parameters within the commercial sector [20].

3.2 HOMER Software Use

However, for this research thesis, use of an alternative software is known as Hybrid Optimization of Multiple Energy Resources (HOMER) software. This software offers the capability to analyze renewable energy data across diverse locations. Numerous locations whether commercialization and residential areas, this thesis is based on simulation for the small community (NUST University, Islamabad) regarding of the university's study.

The HOMER Software comprehends a wide range of features in simulation, including the ability to forecast on solar irradiance system and the wind power efficiency and outputs, as well as the integration of converter storage cells' devices for storing electricity for the future use. The optimization scenarios and sensitive simulation parameters of these renewable energy systems facilitate the financial cost analysis by the buyers and the prediction of the future circumstances, offering a significant advantage.

For this study on renewable hybrid microgrids in replication analysis, HOMER Pro software is used for further cases in these small areas. Developed by the National Renewable Energy Laboratory (NREL), this software serves as a comprehensive techno-economic modeling tool. It plays a pivotal role in optimizing this design and implementation of hybrid microgrids. It also considers the net present costs of utilities in terms of renewable energy. Furthermore, it conducts sensitivity analyses on uncertain variables, providing valuable insights into the resilience and cost-effectiveness of the designed microgrid simulation systems.

CHAPTER 4: METHODOLOGY

4.1 Methodology Cases

The methodology engaged in this research thesis, titled "Evaluation of Techno-Economic Design and Implementation of Solar-Wind Hybrid Microgrid System for National University of Sciences and Technology (NUST), Islamabad (locate in images in Figure 4.1 and Figure 4.2 presents a structured approach to investigating these renewable energy systems through simulation, their functionality, and the financial implications of their optimization. It specifically emphasizes on Solar PV, Wind Power turbines, National Grid, and storage of electricity cells for generated electricity. It is also emphasis on technical and economic feasibility cost using the HOMER simulation for analysis and critical conclusion on this University's community.

The research structure begins with comprehensive literature reviews, as detailed in Chapter 2: Literature Review, aimed at understanding the current trends of this area and the state of research in hybrid renewable energy systems (Microgrid) studies. Subsequently, this case is further analyzed using HOMER simulate application (Chapter 3: Simulation Setup throughout the advanced computer applications for prediction in the forecasting periods. Solar PV and wind power units, their data, measured in kilowatts (kW), are collected during the simulation stage, including the solar irradiance and wind speed charts specific to the University region in Islamabad. In fact, these economic features are depicted in charts and graphs corresponding to the renewable and conventional energy's electricity throughout annual year.

Primary data collection such as energy consumption patterns, with appropriate solar PV panels, wind turbines, grid utility and storage cell models are chosen to be efficient based on the energy needs and environmental conditions of this small region. These data are stored continuously on the computation machineries, considering various environmental simulation factors such as hot weather, rainy seasons, or cold temperatures

in this University community. Configuration of the hybrid system requires parameters to be specified for their performance.

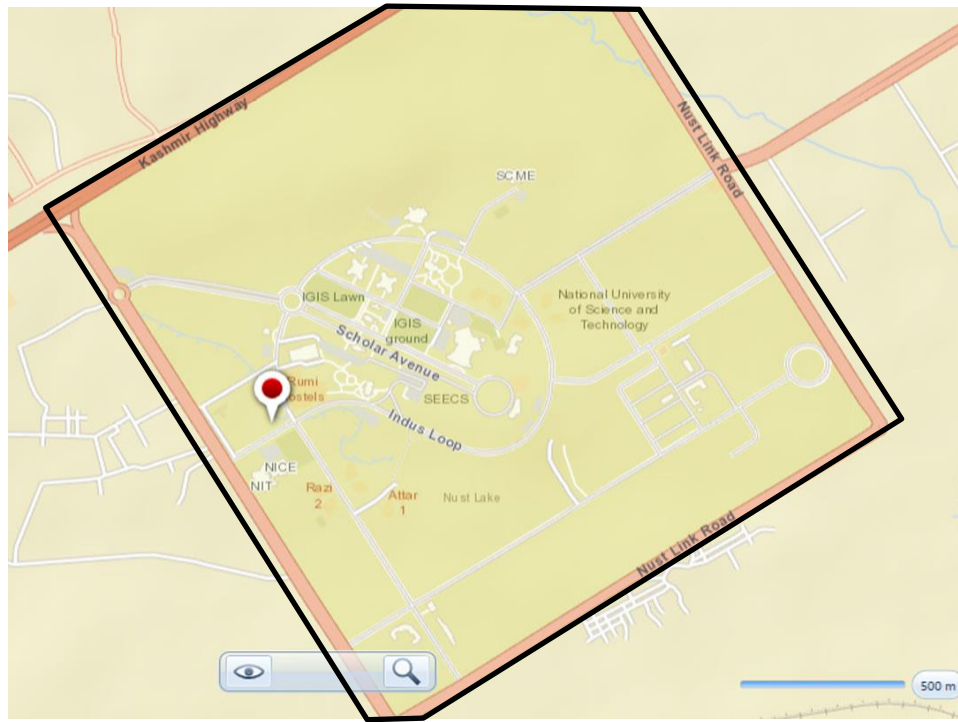


Figure 4.1: University Community displayed by HOMER Simulation



**Figure 4.2: Small Community (University), Islamabad
Satellite View (Source: [Suparco](#))**

Table 4.1: University Community based HOMER Software

University Community Rate	Value
Discount rate (%):	22.00
Inflation rate (%):	25.00
Annual capacity shortage (%):	1.00
Project lifetime (years):	25.00

The HOMER simulation uses specific values for the University community's rate details, automatically identifying its location (as shown in Figure 4.1). These values, released by the State Bank of Pakistan in March 2024, include current and projected rates (above Table 4.1). The discount rate, which discounts future cash flows to their present value, is 22.00% as of March 2024. This rate, adjusted to manage monetary policy, may change over time. The inflation rate, reflecting the general price increase for goods and services, is 25.00% (March 2024). The project lifetime for this community's components is set at 25 years. These factors significantly influence the energy data and projections used in HOMER software.

For Solar PV, the selected model is Trina Tallmax M plus, with a rated capacity of 0.345 kW, a flat plate panel type, 72-cell monocrystalline module, and a maximum efficiency of 17.80%. Stakeholders are advised to invest at least US\$174.30 as capital, replacement cost is US\$143.48 annually, with additional operational and maintenance costs of US\$1.74 ([Forbes Currency Converter](#) – 24th July 2024) per year, over a duration of 25 years, with a de-rating factor of 80.00%. The details are displayed in the Solar PV Trina Tallmax M Plus table in Chapter 5: Results and Discussion. One part of the solar PV installation in the University of the small region is shown in Figure 4.3. It shows the functionality of the solar PV being collected by DC current from the extremely hot air temperature environment, ideally to transform electrons movement from hot to less hot in the surface of the solar PVs' cells. This behavior can happen in the afternoon every day.

These PV collected units are stored afterwards in the storage cells' devices for the usage at present and the future duration.



Figure 4.3: Solar PV Site Installation

Regarding the Wind Power Turbine, the chosen model is AWS HC 1.5 kW Wind Turbine, with a rated capacity of 1.5 kW, a 3.2-meter rotor diameter, and a lifetime of 20 years. Stakeholders should invest US\$322.83 ([Forbes Currency Converter](#) – 24th July 2024) per year for its capital cost, with replacement and operational and maintenance costs of US\$251.09 and US\$32.28 respectively. The table is presented in the following chapter. The wind power device for this university is displayed in Figure 4.4 (below). The name of this wind turbine is AEOLOS. AEOLOS devices are manufacture by the United Kingdom,

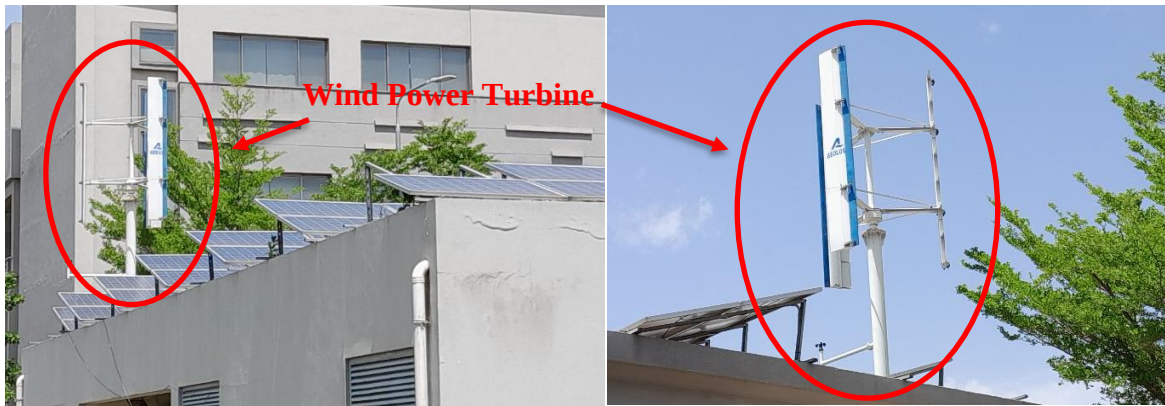


Figure 4.4: AEOLOS Wind Turbine

producing the renewable generation products. This device can rotate like a wind power fan turbine with the mechanical brake system.

The conventional energy source for NUST University is a Genset component, which is used for fossil fuel consumption to generate electricity. As illustrated in Figure 4.5 (below), this conventional component is located diagonally to the USPCAS-E Campus at NUST University, Islamabad. The Genset component has fuel curve slope and intercept, toxic gasses emissions and fuel details as mentioned in Table 5.5. The capital cost per kilowatt (kW) for this component is US\$258.26 ([Forbes Currency Converter](#) – 24th July 2024), and the replacement cost per kW is US\$215.22. Additionally, the operational and maintenance cost per operating hour and per kW is US\$0.04. The fossil fuel price per liter for this university is US\$0.99, with an expected lifetime of 40,000 hours.



Figure 4.5: Genset Conventional Supply



Figure 4.6: National Grid Transmission (132 kV NUST Substation)

The grid connected to the University's parameters, as depicted in Figure 4.6 (above), is linked to the national electric grid supplied by Islamabad Electric Supply Company (IESCO), Pakistan. This national supply primarily relies on several electricity production factors, mostly from hydroelectric power plant (water dams); a renewable energy source. The substation at NUST University typically reduces the high voltage from the grid utility to a medium/low voltage level suitable for consumer use (230/400 VAC units). The given data simulation has the grid power price is US\$0.16 per kWh ([Forbes Currency Converter](#) – 24th July 2024). However, there is no sell-back price for the University's electric load concerning the national grid electricity, contrasting other renewable energy sources at NUST Islamabad such as solar PVs and wind power. This simulation is set to a simple electric rate without net metering for power return to the grid. In fact, the electricity generated by NUST University cannot be fed back into the national grid because the IESCO distribution policies do not allow for NUST's net metering due to grid utility policies and technical constraints. The grid utility electricity load in the simulation is 20,000 kW (20 MW) to NUST University. Additionally, there are greenhouse gas emissions associated with the grid utility, as shown in Table 4.2 below. The simulation indicates emission rates for Carbon Dioxide and Sulfur Dioxide (toxic matter) at 632 g/kWh and 2.74 g/kWh, respectively. These are average emission values for the university, but there is still a need to reduce them further.

Table 4.2: Grid Utility Greenhouse Emission

Greenhouse Gas	Value
Carbon Dioxide (g/kWh)	632.00
Sulphur Dioxide (g/kWh)	2.74
Nitrogen Oxides (g/kWh)	1.34

Before any excess electricity can be stored in the storage devices, a converter needs to be installed at the NUST installation site. This converter system converts AC output currents into controlled DC currents using rectifier technology. The given simulation has the capital cost for this converter system is US\$38.77 ([Forbes Currency Converter](#) – 24th July 2024), and the replacement cost is US\$28.00. There is no operation and maintenance cost associated with this system and has a lifespan of 15 years. In addition, the efficiency of the inverter (DC conversion to controlled AC) and rectifier (AC conversion to controlled DC) is 95 percent each.

Once completed the phase of converter system, next one is to store electricity in storage devices. The selected storage energy cells are the Generic 1 kWh Lithium-Ion (Li-Ion) model. The specifications include a nominal voltage of 6V and a nominal capacity of 1 kWh. The nominal capacity of this lithium-ion battery is 167Ah (Ampere Hour) and the roundtrip efficiency is 90 %. So, the maximum charge current in Amperes is 167 A and the maximum discharge current is 500 A at this University. This storage device consists of one unit with a string size of 6 volts, resulting in a minimum state of charge of 20%. The capital cost for this model is US\$197.45 ([Forbes Currency Converter](#) – 24th July 2024), while the replacement cost is US\$179.50. Additionally, the operational and maintenance cost is US\$0.04 per year, with a lifespan of 15 years.

HOMER Software can analyze resource availability, load requirements, and system components, running simulations to assess performance under varying conditions. Optimization features can identify cost-effective results and efficient system designs, estimating costs associated with installation, operation, and maintenance.

Furthermore, sensitivity analysis is conducted to understand how changes in input parameters can affect system performance and cost. The proposal regarding the evaluation of microgrid system is evaluated by assessing these potential environmental benefits, such

as reductions in greenhouse gas emissions resulting from the implementation of the renewable energy system.

CHAPTER 5: RESULTS AND DISCUSSIONS

As per the analysis conducted throughout this research study stage and using the HOMER software as its main tool for this thesis, Figure 5.1 illustrates the schematic design of the microgrid hybrid system implemented in the University community, Islamabad. In this schematic design, the generation sector – primarily relying on conventional energy sources, transmission utility grid and the wind power source are interconnected with the main electric hub through the AC method. The Alternating Current (AC) sinusoidal waves can absorb electricity much quicker within short timing. Meanwhile, the solar PV system paths are having its renewable energy output through the Direct Current (DC) method. Both AC and DC currents are then stored in Battery storage or battery energy storage systems (BESS) for the future electricity consumption. Figure 5.2 is containing solar PV, wind power and BESS pathways. The converter block plays an essential role in converting AC energy into DC form and facilitating its transfer to the storage system.

Given that the renewable energy generated may not consistently meet the electricity demands of the area's facilities, conventional energy sourced from generators is integrated into the schematic system. The output of renewable energy is reliant upon prevailing weather conditions, relying on inputs from meteorological weather centers in real-time. The final phase (below) depicted in Figure 5.1 pertains to the cost sectors, which holds significant importance for the further development and implementation costs of renewable energy solutions.

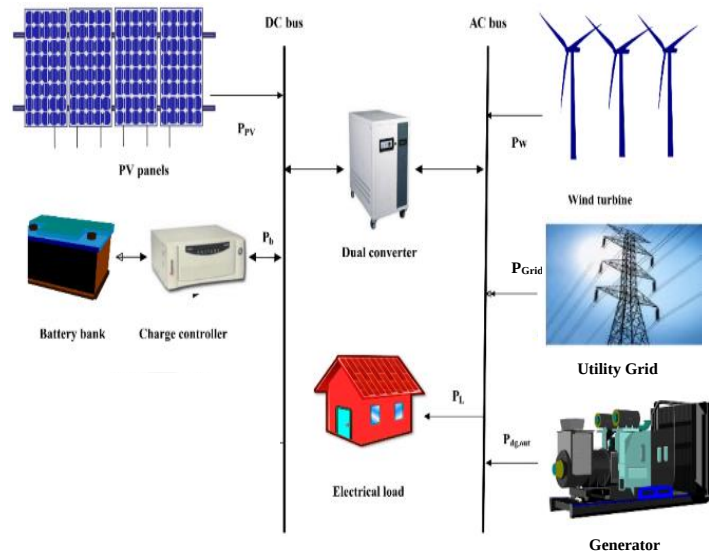


Figure 5.1: Smart Microgrid Hybrid System

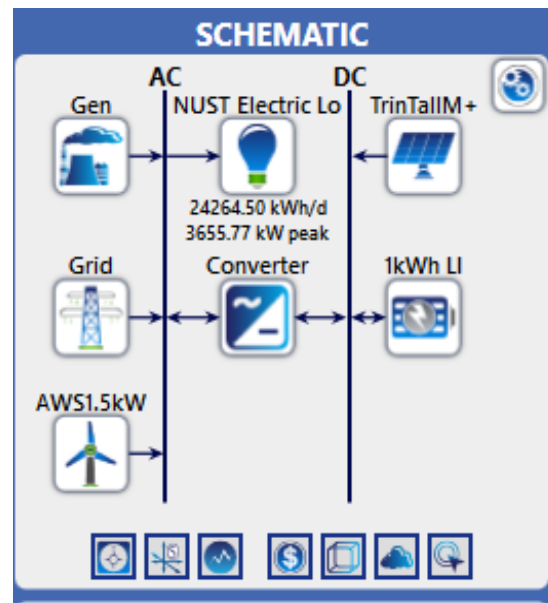


Figure 5.2: HOMER Schematic for Hybrid Microgrid

According to Figure 5.2, which follows a similar schematic pattern as Figure 5.1, the methodology illustrates how both AC and DC currents are stored in designated storage areas. The PV panels generate DC current, supported by DC batteries, while wind power, transmission lines and the generator produce AC currents. The charge controller, a key storage device, operates exclusively on DC current. Therefore, AC sources must be

converted to controlled DC before being stored. This figure aims to demonstrate that renewable energy sources can produce current; however, their efficiency is significantly lower compared to conventional energy sources.

Table 5.1: Converter Control

Component	Min Qty	Max Qty	Bus
Generator	0	20	AC or DC
Storage	0	10	DC
PV	0	10	AC or DC
Wind Turbine	0	2	AC or DC
Converter	0	1	AC or DC
Boiler	0	1	Thermal
Hydroelectric	0	1	AC or DC
Hydrokinetic	0	1	AC or DC
Reformer	0	1	Hydrogen
Electrolyzer	0	1	AC or DC
HydrogenTan	0	1	Hydrogen
Grid	0	1	AC
Thermal Load	0	1	AC

Consequently, a microgrid hybrid system is proposed as a solution to balance renewable and conventional energy, enhancing overall power efficiency. The converter shows a vital role in usage both AC and DC currents as its main parameters such as capacity per time and efficiency performance which has been occurrences. It processes components that produce either AC or DC energy output. As such, some of these components are functioning in Table 5.1 (above). The generator's maximum quantity is 20 units while the storage cells and the Solar PV are ranking maximum quantity of 10 units in this small area. The Wind Power maximum quantity is 2 and maximum 1 unit occurs in the rest of these renewable components.

Therefore, this table corresponding to the given simulation aligns with the microgrid hybrid system in this University, enabling the optimization of power output from combined solar PV and wind power sources. This system aims to reduce reliance on conventional generators and thus lower energy costs to a feasible range. The technical aspects of the components listed in Table 5.1 are designing an effective microgrid hybrid system. This system balances renewable energy (solar PV and wind power) with conventional energy (fossil fuel generators) to achieve optimal power output and costs.

By integrating renewable energy, this system improves output efficiency and hence reduces the need for fossil fuel input. Consequently, the University community in Islamabad can minimize the operating hours of fossil fuel generators. Additionally, the forecast of energy production by the microgrid system can be balanced both visually and data-wise, ensuring efficient management of energy resources. This approach not only enhances energy sustainability but also contributes to lowering the carbon footprint of the University community.

5.1 Electric Load for University Community

The NUST Electric Load contains 24264.50 kWh/day in this University, Islamabad. The maximum point or else peak value for the electricity output is 3655.77 kW as described in the schematic block. In this load block, there are numerous diagrams connected to the main grid. Subsequently, these diagrams are connected to the electric loads. There are: bar chart, whisker box plot, electric load data per month, yearly

Table 5.2: Nominal Electric Load Summary

Hour	Electric Load (kW) - January	Electric Load (kW) – Weekdays	Electric Load (kW) – Weekend
0	2400.00	2400.00	2400.00
1	2400.00	2400.00	2400.00
2	2400.00	2400.00	2400.00
3	2400.00	2400.00	2400.00
4	2400.00	2400.00	2400.00
5	2400.00	2400.00	2400.00
6	2400.00	2400.00	2400.00
7	8000.00	8000.00	6000.00
8	16000.00	16000.00	8000.00
9	16000.00	16000.00	8000.00
10	16000.00	16000.00	8000.00
11	16000.00	16000.00	8000.00
12	16000.00	16000.00	8000.00
13	16000.00	16000.00	8000.00
14	16000.00	16000.00	8000.00
15	16000.00	16000.00	8000.00
16	16000.00	16000.00	8000.00
17	12800.00	12800.00	6800.00
18	10400.00	10400.00	10400.00
19	8000.00	8000.00	5000.00
20	2400.00	2400.00	2400.00
21	2400.00	2400.00	2400.00
22	2400.00	2400.00	2400.00
23	2400.00	2400.00	2400.00

visualization per electric load and efficiency of electric load include lifetime and capital cost.

This electric load component for this University depicts the values in kilowatts (kW) against the hours during the daily period, starting from January each year. Table 5.2 (above) presents a summary of these values, distinguishing between weekdays and weekends (Referred to Appendix 1 and Appendix 2) in the university's electric output functions. The peak value in electric load supply occurs between hours 8 and 16, indicating that from 8 o'clock morning to 4 o'clock in the evening, there is the highest amount of

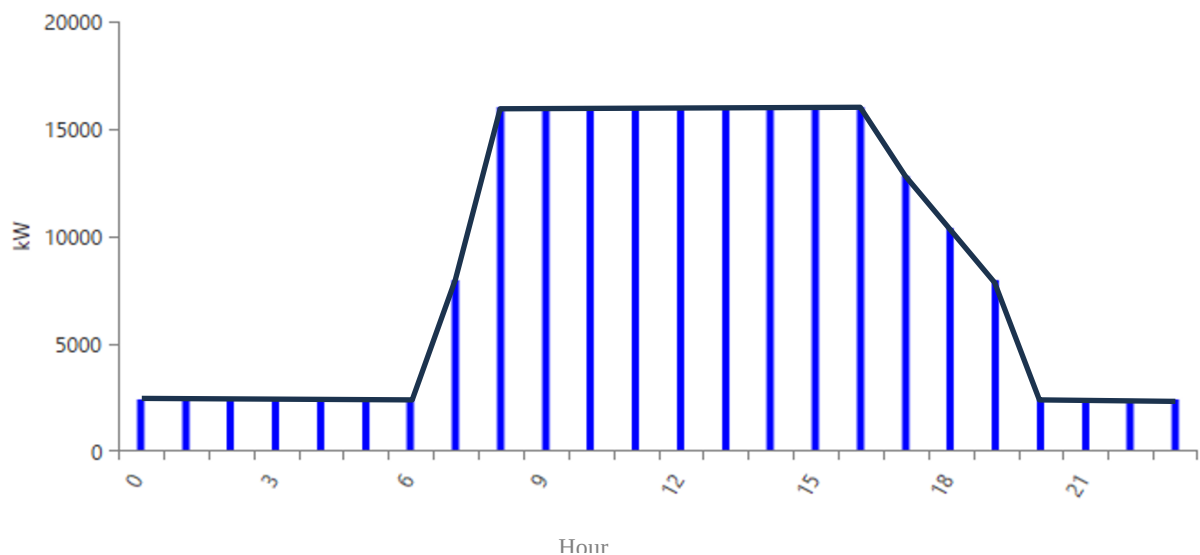


Figure 5.3: Nominal Electric Load

electric load supply throughout both weekdays (16000 kW) and weekends (8000 kW). Referring to Table 5.2 (above), the lowest electric load occurs between 12 AM and 6 AM, then gradually increases. From 8 PM to 12 AM the next day, the load decreases again. Specifically, the lowest load, 2400 kW, is maintained from midnight to 6 AM. The load then gradually rises, reaching 8000 kW at 7 AM, and peaks at 16000 kW between 8 AM and 4 PM. From 4 PM to 7 PM, the load decreases from 12800 kW to 8000 kW. Finally, it drops back to 2400 kW from 8 PM until midnight.

The daily profile of NUST Nominal Electric Load is depicted in a bar chart (Figure 5.3 – above) similar concept for Table 5.2, where the units are represented in kW and hours along the y and x axes, respectively. In this small community (University) of Islamabad,

the highest peak in the bar chart, indicating electric load output, occurs around 8 o'clock morning to 4 o'clock in the evening, reaching almost 16000 kW daily. On the contrary, the lowest point, approximately 2400 kW, is observed between 12 o'clock midnight to 6 o'clock in the morning and between 8 o'clock evening till following 12 AM midnight. Consequently, there is a significant increase and decrease in kilowatts per hours during the morning and evening period respectively. Hence this type of fluctuate can be such balanced by electric load bar chart data to maintain overall neutrality. The bar chart initially shows lower values in 2400 kW until increasing at 7 o'clock in the morning to 8000 kW. Afterwards, it reaches the maximum peak value of 16000 kW between 8 o'clock in the morning and 4 o'clock in the evening. Next, the electric load decline gradually between 5 o'clock to 7 o'clock evening as 16000 kW $\rightarrow$ 12800 kW $\rightarrow$ 10400 kW $\rightarrow$ 8000 kW respectively. It then keeps on horizontal level between 8 o'clock to midnight as its lowest power output as 2400 kW. Hence, these insights highlight the importance of managing electricity supply to meet peak demands efficiently while ensuring energy stability during low-demand periods. Balancing these variations with other electric load data is crucial for maintaining a stable and reliable power supply, thereby enhancing overall energy efficiency and sustainability in the University community.

A whisker box plot (a graphical method of displaying variation in a set of data) is presented representing the load in kilowatts (kW) on the y-axis and the monthly duration on the x-axis at the University, Islamabad. This box plot (Figure 5.4 – below) illustrates various statistical measures, including the minimum (lowest range), first quartile (Q1 - 25th percentile), median (Q2 - 50th percentile), third quartile (Q3 - 75th percentile), and the maximum range (highest value). For instance, the seasonal profile shown in November indicates a minimum value of slightly above 0 kW, a Q1 value of approximately 1000 kW, a median value close to 7000 kW, a Q3 value of approximately 18000 kW, and a maximum value of roughly 24000 kW.

Similarly, the month of January has the minimum, Q1, Q2, Q3, and maximum values for the electric load supply are approximately 500 kW, 2000 kW, 8000 kW, 18000 kW, and 25000 kW, correspondingly. Consequently, the electric loads for other months can be approximated values using Table 5.3 (below) based on Figure 5.4, providing forecasts for these values in the small region location.

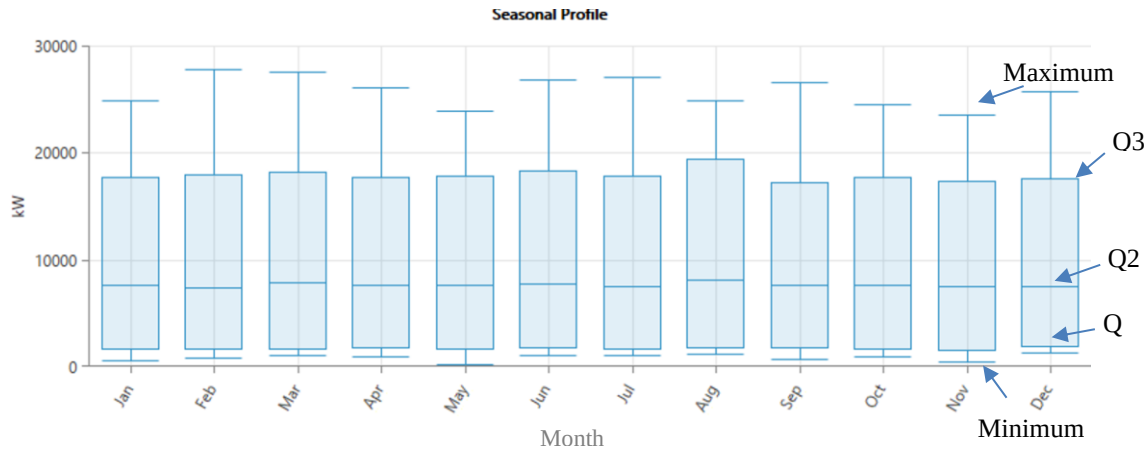


Figure 5.4: Box Plot Electric Load

Table 5.3: Average Generic Electric Load

Month	KiloWatt (kW) - Approximately				
	Minimum	Q1	Q2	Q3	Maximum
January	1000.00	2000.00	7500.00	17500.00	25000.00
February	1250.00	2000.00	7400.00	17600.00	27800.00
March	1500.00	2000.00	7600.00	18000.00	27600.00
April	1400.00	2100.00	7200.00	17800.00	26000.00
May	200.00	1800.00	7200.00	17800.00	24000.00
June	1600.00	2100.00	7400.00	18200.00	26800.00
July	1600.00	2000.00	7500.00	17800.00	27200.00
August	1600.00	2100.00	8000.00	19500.00	24800.00
September	1000.00	2000.00	7500.00	17300.00	26800.00
October	1600.00	2000.00	7500.00	17800.00	24800.00
November	1000.00	2000.00	7500.00	17500.00	24000.00
December	1700.00	2100.00	7400.00	17500.00	25500.00

- **Highest Minimum** :1700 kW at December
- **Highest 25thPercentile** :2100 kW at April, June, August and December
- **Highest 50thPercentile** :8000 kW at August

- **Highest 75thPercentile** : 19500 kW at August
- **Highest Maximum** : 27800 kW at February
- ✓ **Lowest Minimum** :200 kW at May
- ✓ **Lowest 25thPercentile** : 1800 kW at May
- ✓ **Lowest 50thPercentile** :7200 kW at April and May
- ✓ **Lowest 75thPercentile** :17500 kW at January, February and December
- ✓ **Lowest Maximum** :24000 kW at May and November

January:

Q1 – Minimum :- $2000 - 1000 = 1000$ kW (Increase kW from December)
 Q2 – Q1 :- $7500 - 2000 = 5500$ kW (Increase kW from December)
 Q3 – Q2 :- $17500 - 7500 = 10000$ kW (Decrease kW from December)
 Maximum – Q3 = $25000 - 17500 = 7500$ kW (Decrease kW from December)

February:

Q1 – Minimum :- $2000 - 1250 = 750$ kW (Decrease kW from January)
 Q2 – Q1 :- $7400 - 2000 = 5400$ kW (Decrease kW from January)
 Q3 – Q2 :- $17600 - 7400 = 10200$ kW (Increase kW from January)
 Maximum – Q3 = $27800 - 17600 = 10200$ kW (Decrease kW from January)

March:

Q1 – Minimum :- $2000 - 1500 = 500$ kW (Decrease kW from February)
 Q2 – Q1 :- $7600 - 2000 = 5600$ kW (Increase kW from February)
 Q3 – Q2 :- $18000 - 7600 = 10400$ kW (Increase kW from February)
 Maximum – Q3 = $27600 - 18000 = 9600$ kW (Decrease kW from February)

April:

Q1 – Minimum :- $2100 - 1400 = 700$ kW (Increase kW from March)
 Q2 – Q1 :- $7200 - 2100 = 5100$ kW (Decrease kW from March)
 Q3 – Q2 :- $17800 - 7200 = 10600$ kW (Increase kW from March)
 Maximum – Q3 = $26000 - 17800 = 8200$ kW (Decrease kW from March)

May:

Q1 – Minimum :- $1800 - 200 = 1600$ kW (Increase kW from April)
 Q2 – Q1 :- $7200 - 1800 = 5400$ kW (Decrease kW from April)

Q3 – Q2 :- $17800 - 7200 = 10600$ kW (No Change kW from April)
Maximum – Q3 = $24000 - 17800 = 6200$ kW (Decrease kW from April)

June:

Q1 – Minimum :- $2100 - 1600 = 500$ kW (Decrease kW from May)
Q2 – Q1 :- $7400 - 2100 = 5300$ kW (Increase kW from May)
Q3 – Q2 :- $18200 - 7400 = 10800$ kW (Increase kW from May)
Maximum – Q3 = $26800 - 18200 = 8600$ kW (Increase kW from May)

July:

Q1 – Minimum :- $2000 - 1600 = 400$ kW (Decrease kW from June)
Q2 – Q1 :- $7500 - 2000 = 5500$ kW (Increase kW from June)
Q3 – Q2 :- $17800 - 7500 = 10300$ kW (Decrease kW from June)
Maximum – Q3 = $27200 - 17500 = 9700$ kW (Increase kW from June)

August:

Q1 – Minimum :- $2100 - 1600 = 500$ kW (Increase kW from July)
Q2 – Q1 :- $8000 - 2100 = 5900$ kW (Increase kW from July)
Q3 – Q2 :- $19500 - 8000 = 11500$ kW (Increase kW from July)
Maximum – Q3 = $24800 - 19500 = 5300$ kW (Decrease kW from July)

September:

Q1 – Minimum :- $2000 - 1000 = 1000$ kW (Increase kW from August)
Q2 – Q1 :- $7500 - 2000 = 5500$ kW (Decrease kW from August)
Q3 – Q2 :- $17300 - 7500 = 9800$ kW (Increase kW from August)
Maximum – Q3 = $26800 - 17300 = 9500$ kW (Increase kW from August)

October:

Q1 – Minimum :- $2000 - 1600 = 400$ kW (Decrease kW from September)
Q2 – Q1 :- $7500 - 2000 = 5500$ kW (Decrease kW from September)
Q3 – Q2 :- $17800 - 7500 = 10300$ kW (Increase kW from September)
Maximum – Q3 = $24800 - 17800 = 7000$ kW (Decrease kW from September)

November:

Q1 – Minimum :- $2000 - 1000 = 1000$ kW (Increase kW from October)
Q2 – Q1 :- $7500 - 2000 = 5500$ kW (No Change kW from October)
Q3 – Q2 :- $17500 - 7500 = 10000$ kW (Decrease kW from October)
Maximum – Q3 = $24000 - 17500 = 6500$ kW (Decrease kW from October)

December:

Q1 – Minimum :- $2100 - 1700 = 400$ kW (Decrease kW from November)

$Q2 - Q1 :- 7400 - 2100 = 5300 \text{ kW}$ (Decrease kW from November)
 $Q3 - Q2 :- 17500 - 7400 = 10100 \text{ kW}$ (Increase kW from November)
 $\text{Maximum} - Q3 = 25500 - 17500 = 8000 \text{ kW}$ (Increase kW from November)

Appendix 1 and Appendix 2 present the electric load measurements per hour for weekdays and weekends, respectively, in the small region location. Upon comparison, it is noted that both sets of data are slightly different for maximum peak kW for weekdays and weekends. The weekdays between 8 AM and 4 PM is the highest peak data (16000 kW) daily.

The weekend highest peak value is less for those of weekdays' peak values (8000 kW) between 8 AM and 4 PM compared to 16000 kW on weekdays. Overall, the highest recorded value for electric load in both tables is 16000 kW. On the contrary, the lowest recorded value for electric load, when minimal load is supplied, is 2400 kW.

Referring to the comparison benchmark, the data for electric load on weekdays and weekends in this University community presents significant challenges in achieving accuracy. The data could be either positively accurate or false, making future predictions uncertain. Although the HOMER simulation provides a precise location, as well as accurate inflation and discount rates for Islamabad, Pakistan, it may still display incorrect data. This accuracy level reflects historical electric load consumption data in HOMER simulation calculations, impacting electricity usage in the University. Precision in metering actual electric consumption data is essential to ensure accuracy. On the contrary, collecting and optimizing this data for future use could be an effective approach.

The analysis of weekday period in the University's area regarding on power consumption data exposes numerous patterns which are slightly different on weekend one. During the weekdays peak hours typically 9 hours, the power usage in this University is significantly higher percentage than 9 hours of weekends. On the contrary, during the off-peak hours which start from 6 PM to 6 AM, the power output consumption drops dramatically to lower values.

The weekend analysis of power consumption has slight differences towards weekdays. The Weekdays period as proved above shows predictable peak hours during the University community's timeline such as the offices, campuses and residential activity. However, the weekends show a more gradual decrease in the morning because of Saturday and Sunday are off days compared to the weekdays. According to Season, the summer months for weekends drive a less energy demand like that on weekdays, emphasizing the need for efficient energy management strategies.

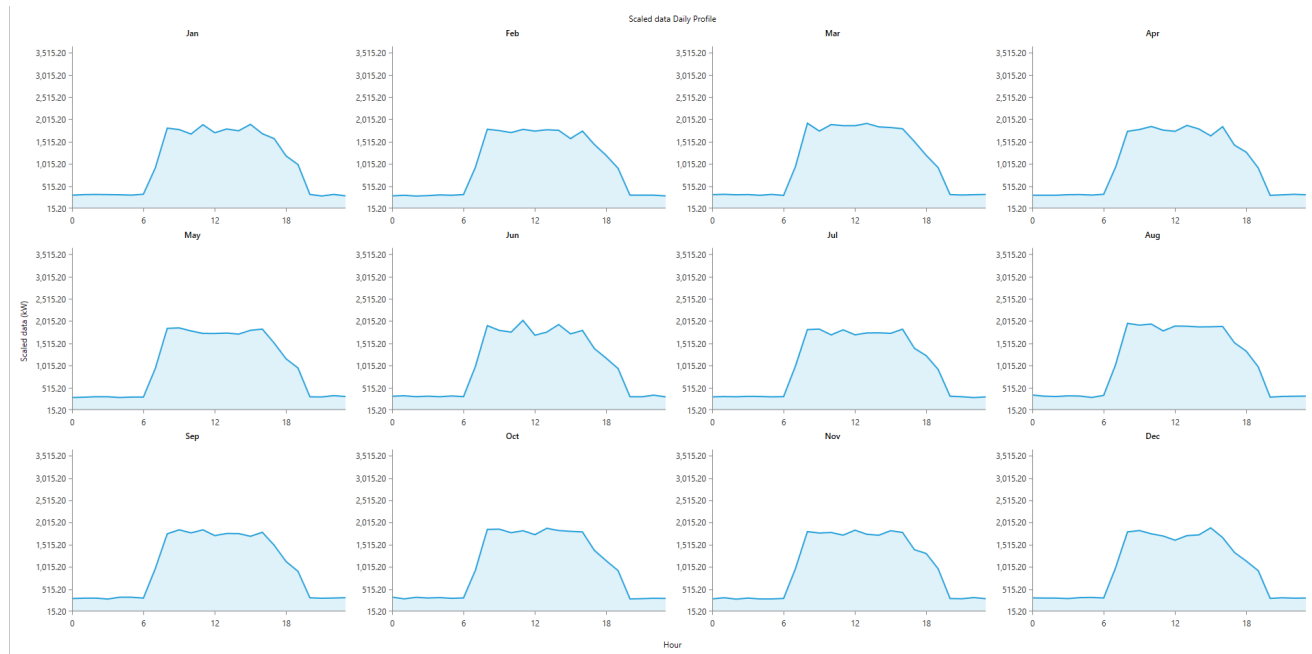


Figure 5.5: Scaled Data Daily Electric Load

Similarly, the tables for the weekends and weekdays data are graphically represented in Figure 5.5 (above and Appendix 4). In each of the twelve graphical representations, the highest electric load for the University consistently peaks between 8 AM and 4 PM monthly with approximately below 2015.20 kW as shown in the figure 5.5. The lowest values occurring is between two ranges: from midnight to 6 AM and from 8 PM to next midnight. This pattern demonstrates that the electric load regularly reaches its maximum kilowatts every month, while a low horizontal curve is depicted during the initial and late hours period.

Hence, the integration of this microgrid hybrid system observed in this University can significantly impact of the electric load patterns. These patterns have function by combining both renewable energy and conventional energy sources and thus having balance system toward achieving a more stable and efficient energy supply. Moreover, this system can store excess energy produced during low-demand periods and hence utilize these power units during peak demand times.

Furthermore, the ability of this microgrid hybrid system is to balance both energy (conventional and renewable) supply and then customers' demand as the main concern which have such benefits, observed in these graphs above. Initially hours, the electric load is at its lowest kW value and thus can store excess renewable energy. If this stored energy by the storage charge devices can be used to meet this higher demand observed in the evening around 6 to 9 PM, then it could reduce the usage on conventional energy sources, reduction of harmful gasses including Carbon Dioxide footprint and minimizing the environmental impact. Furthermore, the peaks observed by the Figure 5.6 – below can be effectively managed by these microgrid hybrid systems in the University, reconfiguration by renewable energy cases only. The Solar PV systems are mostly used around noon if the sunlight is abundant without any circumstances issues such as most pattern of the cloudy skies environment. Moreover, the wind power can support these renewable energy fraction towards becoming the university as renewable standalone electricity demand. By harnessing this peak solar-wind energy, the system can supply a significant portion of the midday electric load and hence enhance energy efficiency. Therefore, the microgrid hybrid system after control for these parameters can be optimized on these utilization of renewable

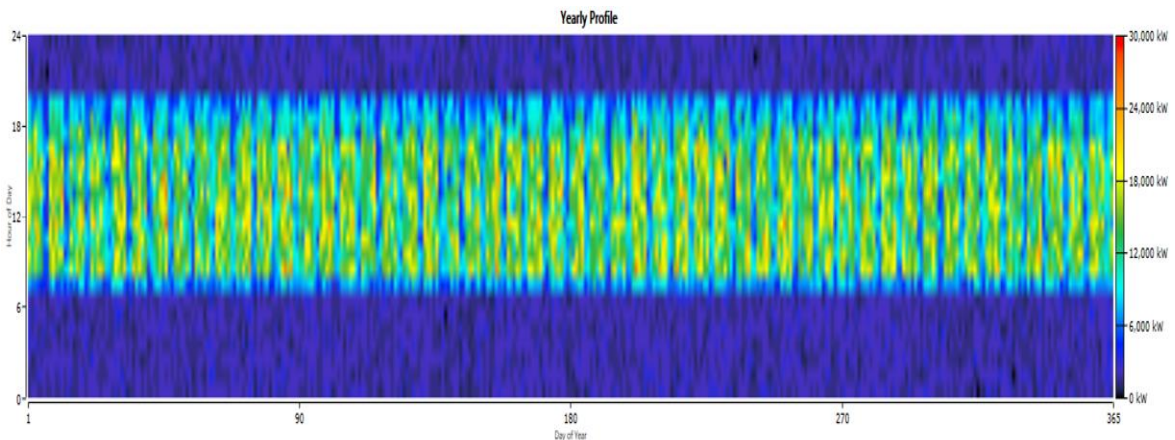


Figure 5.6: Electric Load Visualization

energy sources. It is also reducing its dependence on fossil fuels towards ensuring a stable and reliable power supply and environmental sustainability by minimizing greenhouse gas emissions. These graphical analyses of the electric loads can achieve energy efficiency and stability in this University.

According to seasonally period (Figure 5.6 – above), the summer months like May to September (approximately 150 to 240 days) exhibit such power demand similar with spring (60 to 120 days) and winter (330 to next 30 days) seasons in this University. The winter months similar from December to February show a slight decrease in power demand compared spring and summer seasons. This demand is covered by winter protection such as heat systems and hot water circulations. Hence, these insights of the energy demand between the renewable and conventional forms can result in energy management strategies and promote this energy efficiency furthermore.

The Electric load visualization (depicted in figure 5.6) has interval (right side) for its power output; mostly in colors. The black-dark blue color indicates approximately 0-6000 kW. The highest peak hour between 8 AM and 4 PM shows a significant change from blue color (0-6000 kW) to green-to-yellow-orange color (12000-24000 kW), suggesting that more electricity is being used by the University, Islamabad, during these 9 hours daily. While the dark blue-black color is displayed between midnight and 4 AM, indicated around 0 to less than 6000 kW daily. This visualization shows more details below.

- **Dark blue to blue:** Represents the electric load of 0 – 6000 kW power consumption. This power kW unit is between midnight and 6 AM as well 8 PM to midnight for off-peak hours, consistent in low energy usage.
- **Light blue to cyan to light green:** Indicates between 6000 to 12000 kW power usage, occasionally spikes these power outlets around outline of these 9 hours maximum power output energy.
- **Green to dark yellow:** Corresponds from 12000 kW to 18000 kW power consumption, showing regular appearances on the energy electricity, indicating lesser appearance than yellow color energy consumption.

- **Yellow to orange:** Reflects between 18000 kW to 24000 kW power demand. This power system is rare yet noticeable in this University's efficiency.
- **Dark orange to red:** Signifies this color pattern is between 24000 kW and 30000 kW power usage. However, these power outline supplies in the University are very irregular.

The visualization for this power demand indicates with analysis of power consumption, having numerous yet fascinating patterns. Dark blue-blue (6000 kW) which is dominated in this figure suggesting consistent at low energy usage of the power outlets in this University community. However, there are mostly spikes throughout the years between 6000 kW to 12000 kW (light blue-cyan-light green), indicate almost all periods of increased 9 hours demand. Similar, regular appearances of energy providers between 12000 kW to 18000 kW (green-dark yellow) shows significant power outputs at this university as well as from 18000 kW to 24000 kW (yellow-orange) highlight occasional higher usage. Very infrequently, 24000 kW to 30000 kW (dark orange-red) occurs small stances. So, understanding these patterns can inform energy management strategies in power consumption in the University. The Table 5.4 (below) displayed the electric load metric in scaled (adjustment of energy demand values).

Table 5.4: Electric Load Metric

Metric – NUST Power	Load
Average (kWh/day)	24,264.50
Average (kW)	1011.02
Peak (kW)	3655.77
Load Factor	0.28
Current Load Type	Alternating Current
Scaled Annual Average (kWh/day)	24,264.50

5.2 Generator Provider

The schematic diagram for the generator model in the University community can feature towards the Genset generator, powered by diesel fuel. Table 5.5 (below) provides an overview of the functions and values associated with this Genset generator. Despite the utilization of renewable energy sources in the University, the generator mainly relies on

conventional energy. This is primarily due to the moderate efficiency rating of renewable energy, which contribute less than one hundred percent of the electricity generated daily. Therefore, there is a need for a balanced approach, where fossil fuels are utilized to supplement the electricity supply. However, it is important for the electricity generated from conventional energy sources to meet optimized standards. Consequently, both renewable and conventional energy sources are utilized to meet the electricity needs of this University.

Table 5.5: Genset Generator Details

Function	Values
Fuel	Diesel
Diesel Fuel Price (US\$/L)	0.99
Lower Heating Value (MJ/kg):	43.20
Carbon Content (%):	88.00
Density (kg/m³):	820.00
Sulfur Content (%):	0.40
Fuel curve intercept	60.10 L/hr
Fuel curve slope	0.24 L/hr/kW
Emissions	
CO (g/L fuel)	16.50
Unburned HC (g/L fuel)	0.72
Particulates (g/L fuel)	0.10
Fuel Sulfur to PM (%)	2.20
NO_x (g/L fuel)	15.50
Generator Cost per capacitor kW (US\$)	
Initial Capital	258.73
Replacement	215.61
Operational and Maintenance (US\$/ operational hour)	0.04
Generator Specific	
Minimum Load Ratio (%):	25.00
Heat Recovery Ratio (%):	0.00
Lifetime (Hours):	40000.00
Minimum Runtime (Minutes):	0.00

Genset Diesel Generator has specific values which are relative to the specifications and operational parameters of this diesel generator. These values are factors such as fuel efficiency, emissions, generator cost, and specific operational details. The fuel efficiency according to Table 5.5 has a lower heating value, representative in the energy content of diesel fuel around 43.2 MJ/kg. Fuel curve intercept and slope indicate the fuel consumption rate (liters per hour) which is based on load (Diesel fuel) having 60.1 Liters per Hour of

fuel curve intercept. The emissions gasses such as Carbon Monoxide (CO), Unburned hydrocarbon (HC), Particulates, and NO_x have levels in grams per liter of fuel in these genset generators. Fuel Properties furthermore have Carbon Content indicates the carbon percentage in diesel fuel, usually around 88%. Sulfur content represents the sulfur percentage that is typically low of 0.4%. Generator Cost corresponds have three factors which are Initial Capital, Replacement and Operational, and Maintenance costs, costly US\$258.48 ([Forbes Currency Converter](#) – 24th July 2024), US\$215.40 and US\$0.04 respectively. These generators have 25% in Minimum Load Ratio and 0% in Heat Recovery Ratio. Moreover, the Lifetime (operating hours) of these generators have active for 40000 hours (about 4 and a half years).

The fuel consumption values can be elaborated in relation to the power output using the diagram in Figure 5.7 (below). This figure corresponds to the data presented in Table 5.6 (below), showing the relationship between fuel consumption and power output. It is evident from Figure 5.7 that there is a direct proportional relationship between fuel consumption and power output, as the graph depicts a positive linear trend. Figure 5.8 (below) displays the efficiency percentage corresponding the same power output from this diesel fuel consumption.

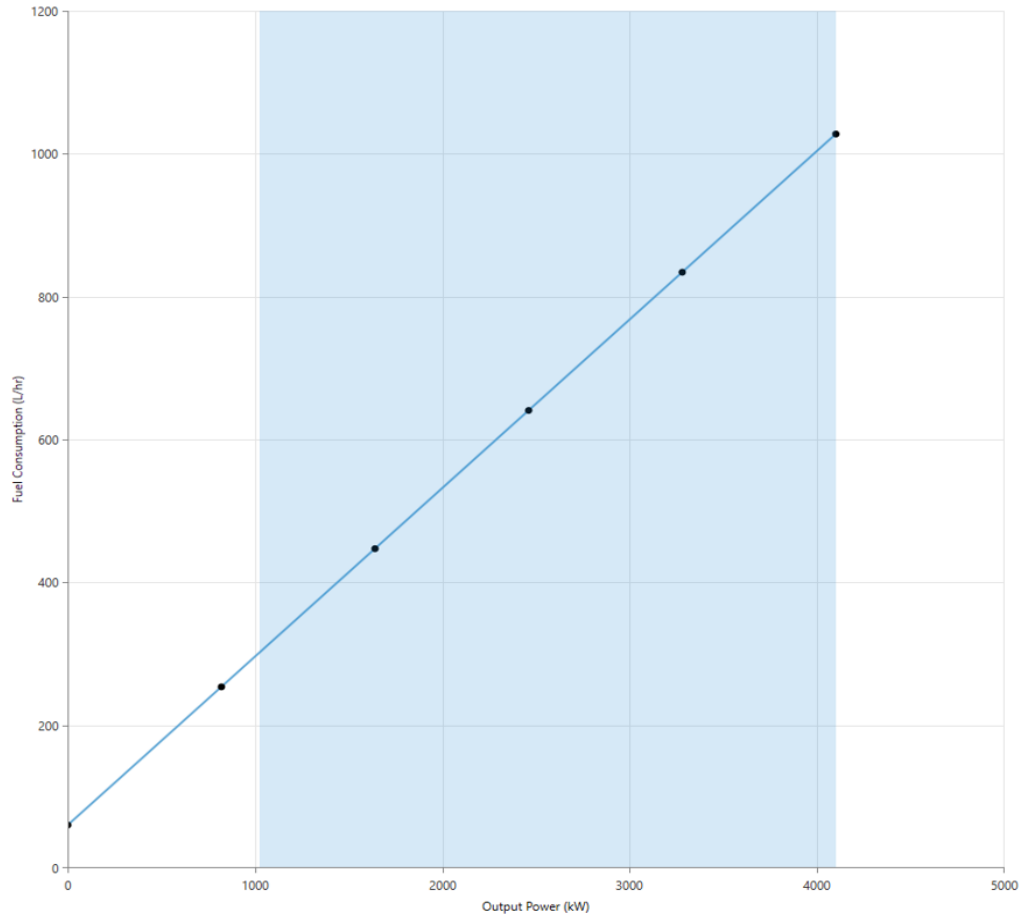


Figure 5.7: Fuel Consumption Versus Power Output Graph

Table 5.6: Fuel Consumption Versus Power Output Data

Output Power (kW)	Fuel Consumption (L/hr)
0	60.14
820	253.64
1640	447.14
2460	640.64
3280	834.14
4100	1027.64

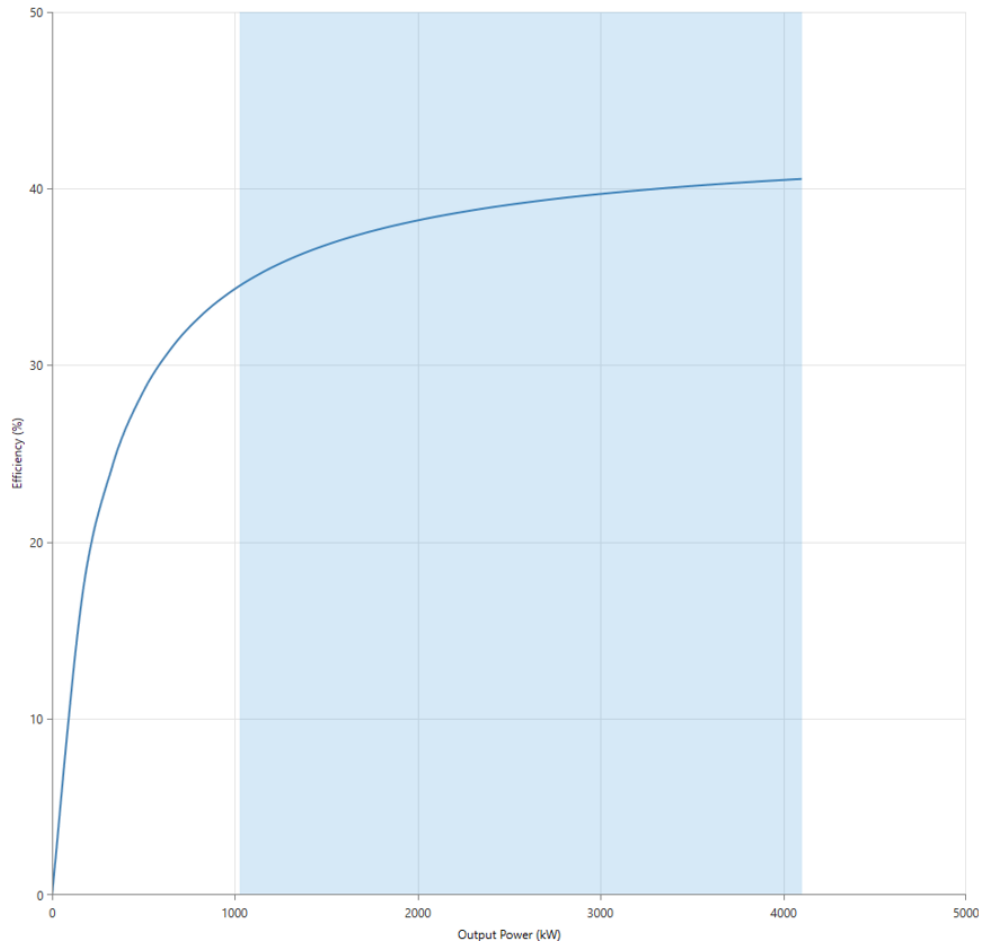


Figure 5.8: Fuel Efficiency Versus Power Output

Hence, the microgrid hybrid system containing renewable energy sources (such as solar, wind, or BESS batteries) with this diesel generator has a better operational behavior with these parameters such as Wind Power system in Table 5.7 and Table 5.8 factors below. The Fuel Efficiency while functioning with microgrid hybrid system has inferred dependency on the Genset because of the alternative power sources. It also has lower operational hours for the Genset, potentially extending its lifespan. This system furthermore via emissions has the rating as lower substance due to this reduced Genset operation. The maintenance with less frequent Genset has usage toward decrease requirements and its cost is relatively reduced fuel consumption too. Hence, the initial capital investment by the stakeholders has been use in these hybrid components in NUST University. As a result, the microgrid hybrid systems promote sustainability, reduce reliance on fossil fuels, and enhance energy resilience.

5.3 Wind Power Turbine Provider

The first renewable energy source that produces energy for the University community, Islamabad is the Wind Power Source. Although, there are no principal components in this community, the university has small wind turbines (AEOLUS-UK Wind Power Vertical Turbine) yet effectively efficiency. The figure for one device out of the wind turbines is mentioned in the Research Methodology section. These small wind powers can ultimately produce the AC current that functions in machineries or else stored it into the storage cell devices afterwards. Since the AEOLUS manufacturer does not mention in HOMER software for simulation, the wind turbine has similar power function too. So, the wind power model is named as AWS HC 1.5 kW Wind Power source, which is 8 meters vertically. Although wind turbines are an excellent source of renewable energy, NUST University has fewer wind power components compared to solar PV installations due to the limited wind currents in this region. Therefore, generators and solar PVs can serve as substitutes for power output when there is insufficient wind at this university location.

Table 5.7: Wind Turbine's Cost Details

Function	Values
Capital Cost (US\$)	323.10
Replacement Cost (US\$)	251.30
Operational and Maintenance Cost (US\$)	32.31
Lifetime (Years)	20.00
Hub Height (m)	8.00

The function of the AWS HC 1.5 kW Wind Power devices, as depicted in Table 5.7 (above), includes the capital cost, replacement cost, and operational and maintenance costs. It also considers the material duration and vertical hub height. The capital cost for installing this renewable energy in this community is US\$323.10 ([Forbes Currency Converter](#) – 24th July 2024). There is also replacement, acquired after the devices surpass their 20-year lifespan, amounting to US\$251.30. The devices, which are installed at a vertical height of 8 meters, require regular maintenance and operational repairs costing US\$32.31 periodically. These expenses are necessary to maintain the microgrid hybrid solutions and reduce the carbon footprint in the University community.

The wind power turbine in the NUST University, Islamabad corresponds to given simulation has a chart displaying the power output from this renewable energy source, with hours on the y-axis and annual time on the x-axis. According to Figure 5.9 (below), the wind turbine produces maximum AC current consistently from 12 PM to slightly after 6 PM throughout the year. The maximum power output varies daily, peaking around day 90 (March) and after day 270 (October), indicated by a yellow color, suggesting a significant increase to roughly between 1200 kW and 4800 kW those days. The initial and final hours in this university show slightly above 0 kW, corresponding to off-peak hours. Despite the color variations in Figure 5.9, the consistent production of renewable energy throughout the day provides a feasible alternative to conventional energy, and the microgrid hybrid system could effectively supply renewable energy to customers efficiently.

The graph for the wind rotor (Figure 5.10) clearly depicts the relationship between

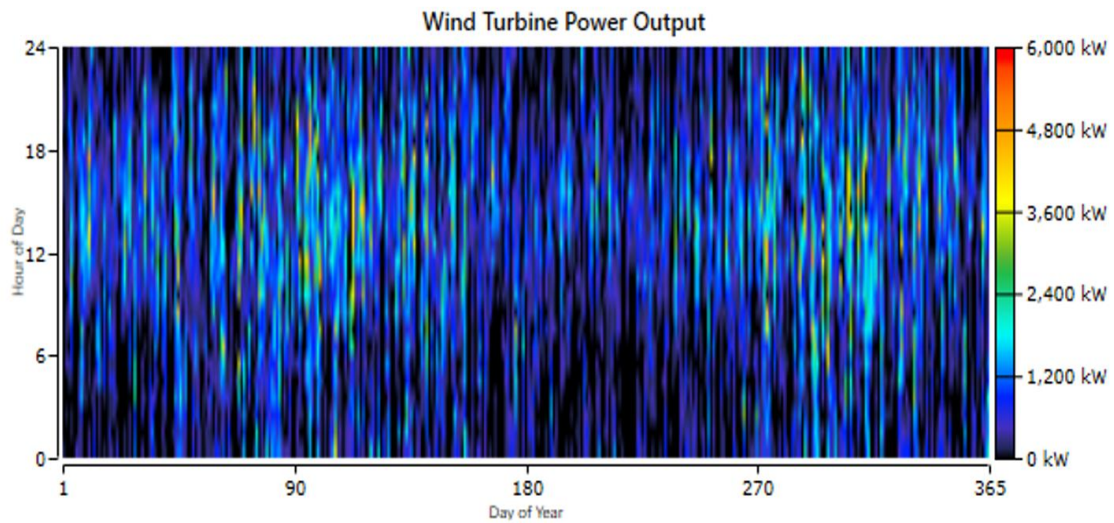


Figure 5.9: Wind Turbine Power Output

power output (kW) and wind speed (m/s) as given simulation in Table 5.8 – below. As wind speed increases, power output also increases. Starting at 2.2 m/s (meters per second), the power output in kW is begin (0.02). When the wind turbine rotates around 8 m/s, this power output reaching approximately 0.68 kW. Beyond this point, the power output rises dramatically, reaching at approximately 1.14 kW speeds when it rotates at 10.2 m/s. Its steeply increases from roughly 10.2 m/s to 12 m/s, achieving a maximum power output of 2.31 kW. According to the wind turbine analysis, the data between wind speed and power

output shown in Table 5.10 mirrors the graph, indicating the efficiency of power output in renewable energy. This data demonstrates the advantages of using a microgrid hybrid system.

The analysis of the Wind Power devices (AWS HC 1.5 kW) demonstrates the importance of managing expenses, including its capital, replacement, and operational maintenance. The wind turbine data (Table 5.10) illustrates steady power production during peak hours regarding the renewable energy feature and highlighting the feasibility of renewable energy as an alternative to conventional energy. The relationship between wind speed and power output further stresses the efficiency of these devices furthermore. Implementing the microgrid hybrid system can enhance energy supply efficiency, reduce carbon or any harmful gasses emissions, and finally ensure a stable power supply for this University community in Islamabad.

This topic is fascinating as it explores alternatives such as renewable energy with less carbon footprints and gas tons in comparison to function with conventional power sources only. However, there are some concerns specific to wind turbines only. On days with less wind regions, the NUST University community may face huge challenges in relying merely on renewable energy microgrid solutions. Additionally, the cost of wind power can be substantial more compared to conventional energy, despite its 20-year lifespan. To address these issues, the microgrid hybrid system could incorporate a generator (conventional fossil fuel system) as a backup solution.

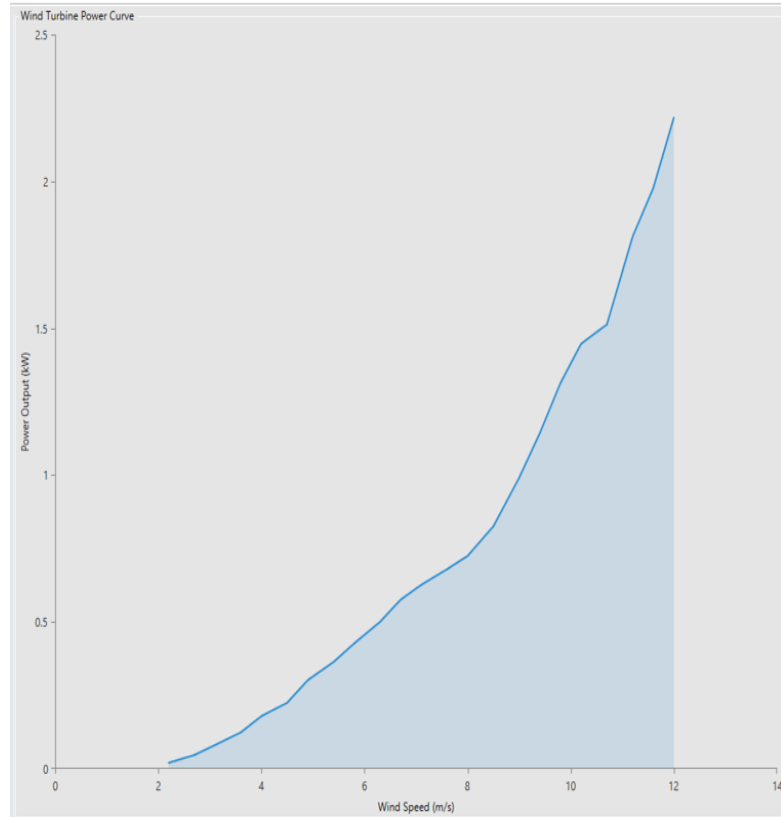


Figure 5.10: Wind Turbine Power Curve

Table 5.8: Wind Speed versus Power Output kilowatts Curve

Wind Speed (m/s)	Power Output (kW)
2.2	0.02
2.7	0.05
3.1	0.08
3.6	0.12
4.0	0.18
4.4	0.22
5.1	0.30
5.6	0.36
6.0	0.43
6.3	0.50
6.9	0.58
7.6	0.63
8.0	0.68
8.4	0.72
9.0	0.83
9.6	0.99
10.2	1.14
11.0	1.39
11.4	1.45
12.0	2.31

5.4 Solar PV Renewable Energy

The second renewable energy source that can be utilized by the University community in Islamabad is solar photovoltaic (PV) devices. The images of these solar cells' surface in the university are depicted in the methodology section. Solar technology in fact can capture energy units from the Sun's rays and keep on collected with peak energy conversion method typically occurring from around 11 AM until sunset during the spring, summer, and autumn seasonal periods. However, the duration of sunlight rays is shorter during the winter season because of Islamabad geographical patterns. The university in the small community uses the Trina Tallmax M Plus solar PV model. Hence, this model is to collect heat-temperature electrons data from the heat rays captured by these cells' surfaces. This energy data, including factors like solar irradiance and on-peak/off-peak solar efficiency, is analyzed using simulation software such as HOMER/RETScreen foundations. The details of the solar PVs are presented in Figure 5.11 (below) and their functions in Table 5.11 below details.

Figure 5.11 is a similar chart to the wind power output shown in Figure 5.10 (above); similar parameters for the power output in kilowatt units. Instead of the blue to red color scale, the solar PV power output is represented in black-red-yellow colors for the power kW units. This chart illustrates the solar irradiance output per hour (Y-axis) and the day period for the year (X-axis). Although the values remain consistent factor, this cost optimization for renewable energy may vary. The graph indicates that the maximum solar

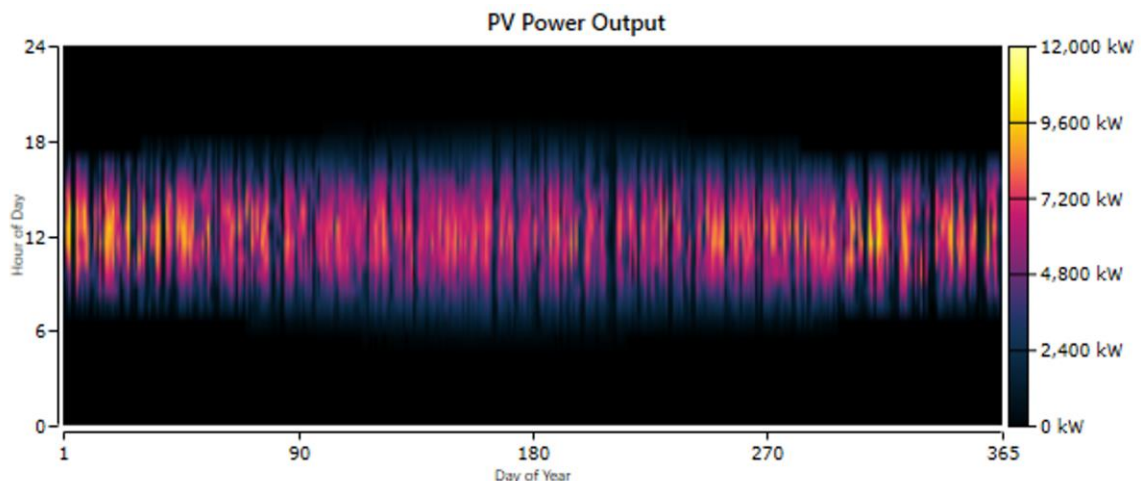


Figure 5.11: Solar PV Power Output

irradiance occurs around the 12th hour (12 PM), with a power output of approximately between 9600 kW and 12000 kW annually. However, it could change to the next year, depending on the geographical patterns in Islamabad.

Given that the year begins in January as an usual setting month pattern, around 180 days (June-July) are affected by the summer conditions. Consequently, the gap for energy between hours should increase vertically axis due to the extremely hot conditions, experienced during this period compared to days 0 (January) to 90 (March) and 270 (September) to 365 (December) days. These 180 days (about 6 months) and nearby particular days showing higher irradiance spots than the start and the end of the year annually, signifying the summer season. However, the yellow color indicates less significant energy in the first 50 (last February) days and the last 60 days (November-December), showing less duration of solar PV power output during these periods. Hence, these bars are way shorter vertically axis compared to the summer season, indicating roughly 6 hours of sunlight rays provided. The estimated solar PV sensors generally produce energy from 8 AM to 5 PM, providing about 9 hours of sunlight annually.

Table 5.9: Trina Tallmax PV Details

Function	Values
Panel Type	Flat Plate
Rated Capacity (kW)	0.345
Material Module	Monocrystalline 72 cell
Power Output Rated (kW)	0.325 – 0.345
Efficiency Maximum (%)	17.80
PV Cost	
Capacity (kW)	1.00
Capital (US\$)	323.28
Replacement (US\$)	251.44
Operational and Maintenance (US\$)	32.33
Lifetime (Years)	25.00
MPPT Lifetime (Years)	15.00
Ground Reflectance (%)	20.00
Tracking System	Horizontal Axis (Monthly Adjustment)

Table 5.9 (above) displays similar factors to the solar irradiance system in Figure 11. The Trina Tallmax PV has a flat surface plate containing 72 solar PV cells

(monocrystalline module) with a rated power output of 0.345 kW. The costs for this solar device are comparable to those for the wind power turbine costs. The capital and replacement costs for the solar device (mentioned in Chapter 4: Methodology) is US\$174.47 and US\$143.60 ([Forbes Currency Converter](#) – 24th July 2024) respectively. The operational and maintenance cost is US\$1.74 for regular repairs. The solar PV has a lifetime duration of 25 years, with a maximum power point tracker (MPPT) efficiency point (maximum solar power efficiency – AM1.5) at 15 years. Therefore, comparing this solar PV renewable energy with wind power as part of a microgrid hybrid system shows significant much better than conventional fuel energy only if these renewable factors are maintained properly.

The University community in Islamabad utilizes two primary renewable energy sources: wind power and solar photovoltaic (PV) devices. Wind power, through the simulation of AWS HC 1.5 kW Wind Power device is function instead of AEOLUS 1.5kW, involves significant costs for installation, replacement, and maintenance factors but provides consistent power output, especially during peak hours. Solar PV devices, specifically the Trina Tallmax M Plus model, also offer a sustainable energy solution with peak efficiency around midday. Both systems require careful cost management and maintenance and also show potential benefits for their effectively efficiency, replacing conventional energy sources as a part of a microgrid hybrid system.

This type of microgrid hybrid system has numerous variations. A few of the statements are implementation of wind and solar PV technologies that highlight a strong commitment to renewable energy. Moreover, the cost analysis for this type of microgrid system provides a realistic view of the financial requirements and long-term benefits in the University community. Data on power output and efficiency can emphasize about the feasibility costs and optimal power output of these renewable energy sources in the local context.

However, there are numerous concerns during the installation of solar PV systems. Stakeholders, particularly investors, have inquiries regarding strategies to ensure the consistent maintenance and operational efficiency of these systems in the University. One

concern is how cloud formulation cover could impact the University community's power supply, potentially reducing the effectiveness of the solar PV power output systems. In such cases, the solar PV systems may still provide some power, but not enough to meet the community's needs fully. Therefore, the microgrid system should consider investigating in these backup solutions, such as optimum fossil fuels and grid utility, to ensure a reliable electricity supply when renewable energy contributions are insufficient.

5.5 Storage Section in University Community

The storage of energy produced by renewable sources such as Solar, Wind and hydro-electricity producer is managed through various modules comprising batteries (DC current) and inverters within the University community's infrastructure. The DC current, such as that generated by solar PVs, can be directed to the storage section. If the wind power turbine processes through AC current, then this AC current will go to the converter control system and afterwards rectification it and finally transformed into controlled DC power. This controlled DC power is then stored in the designated storage sections as well. Once both sectors of renewable energy (referred to hybrid microgrid) generate AC power output through converter controller, it can be readily utilized for various applications in this community in the future times. Examples of appliances are air conditioning, heating systems, and running fans. Furthermore, any surplus power units can be fed back by NUST University into the grid towards the IESCO distribution sector and later be usage to other customers even though there are such issues in terms of the grid utility power policies. It can be also stored in the storage facilities in this university and to use it in the future period. While storing electricity, this charge can present challenges and hence involves in significant costs towards this infrastructure of energy. It also provides invaluable factors during disruptions such as power outages, load shedding electric faults and transient currents. The storage section, essentially functioning as the battery, stores electricity for later use and supplies power to other areas as required. For this purpose, the University in Islamabad can have service in the Generic 1 kWh Lithium-Ion (Li-Ion) storage model. The specifications and calculations associated with this model are elaborated below:

Table 5.10: Generic Li-Ion Details

Function	Values
Nominal Voltage (V)	6.00
Nominal Capacity (kWh)	1.00
Maximum Capacity (Ah)	167.00
Roundtrip efficiency (%)	90.00
Maximum Charge Current (A)	167.00
Maximum Discharge Current (A)	500.00
Generic Li-Ion Cost	
Capital (US\$)	197.66
Replacement (US\$)	179.69
Operational and Maintenance (US\$)	0.04
Lifetime	
Throughput (kWh)	1.00
Time (Years)	15.00
Generic Li-Ion Input	
String Size (Factor)	1.00
Total Voltage (Factor * Nominal Voltage)	6.00 V
Initial State of Charge (%)	100.00
Minimum State of Charge (%)	20.00

The function of Lithium Ion storage devices (Table 5.10 – above) involves the charge and discharge of DC currents to be used in various appliances, as well as their voltage and capacity. The nominal voltage for this type of storage battery is 6 volts, with a capacity of 1 kWh. The maximum capacity for this battery is 167 Ampere Hour (Ah). The maximum electron charge current is 167 Amperes (A) and a discharge current is 500 A. The cost for this generic Li-Ion battery storage is capital: US\$197.45 ([Forbes Currency Converter](#) – 24th July 2024), replacement: US\$179.50, and operational and maintenance: US\$0.04. The lifetime of this storage capacity is 15 years, with a throughput of 1 kWh. The initial input charge of this storage battery is 100 percent, with a minimum charge state of 20 percent, indicating that it can charge the current storage depending on the delivery rate of storage current and its voltage. The total nominal voltage is 6 volts, derived from multiplying the string size by the nominal voltage (string size: 6 x 1V = 6V). Hence, the Generic Lithium-Ion storage systems are essential for managing the charge and discharge of DC currents. The charge and discharge rates of this battery indicate that it can proficiently handle this renewable energy demands for integrating into the microgrid

hybrid system. The costs for maintenance and operational aspect have a sufficient repairs requirement because of the high standard of this battery quality which is beneficial for a long-term usage (15 Years).

However, this battery storage needs to perform high functions to have continuously charge/discharge of currents while in the University's microgrid hybrid system. It also has such energy storage solutions in terms of efficiency and cost-effectiveness when using within this microgrid hybrid system regarding on renewable energy sources. Furthermore, this battery storage has potential drawbacks or limitations when functioning in this microgrid hybrid solution. This behavior is also limited usage for the storage's lifecycle and replacement cost of the impact in this economic feasibility of its integration into renewable energy systems. Thus, the microgrid hybrid system should have optimum solutions with conventional energy as a balance with renewable energy to supply electricity to this community.

5.6 Utility Grid

The national grid of Pakistan supplies electricity to NUST University through various sources of power generation. Hydroelectricity is a major contributor, providing a significant portion of the country's energy demand. For example, Figure 5.12 has a hydroelectric dam in Azad Kashmir, Pakistan where water flow is converted into electricity. This generated electricity is then transmitted throughout high-voltage lines by step-up transformers. This high voltage electricity can transport to the cities, urban centers, rural areas, and villages across Pakistan throughout utility grid transmission lines. Hence, the NUST University utility grid details throughout the months are given in Table 5.11 below.



Figure 5.12: Neelum-Jhelum Hydroelectricity Plant

Table 5.11: NUST Transmission Costs

Month	Energy Purchased (kWh)	Energy Sold (kWh)	Net Energy Purchased (kWh)	Peak Load (kW)	Energy Charge (US\$)
January	161,482.00	488,698.00	-327,216.00	2,640.00	26112.59
February	92,690.00	516,885.00	-424,195.00	2,055.00	14988.50
March	90,288.00	686,979.00	-596,691.00	1,780.00	14600.09
April	57,583.00	848,017.00	-790,433.00	1,560.00	9311.54
May	68,900.00	764,119.00	-695,219.00	1,463.00	11141.59
June	66,643.00	642,314.00	-575,671.00	1,130.00	10776.64
July	87,532.00	547,182.00	-459,650.00	1,681.00	14154.47
August	150,189.00	444,829.00	-294,641.00	2,267.00	24286.39
September	101,534.00	556,268.00	-454,734.00	2,380.00	16418.74
October	76,207.00	828,643.00	-752,437.00	2,040.00	12323.06
November	104,535.00	719,071.00	-614,537.00	1,892.00	16903.89
December	122,520.00	487,915.00	-365,395.00	1,905.00	19812.21
Annual	1,180,102.00	7,530,919.00	-6,350,816.00	2,640.00	190829.72

The energy purchased factor represents the amount of electricity NUST University Islamabad purchased from the national grid, measured in kWh according to the given simulation. The highest energy purchase buyers occur in January (winter season), while the lowest band is in April. Energy sold is the electricity factor where NUST university sells back to the national grid, although no physical implementation currently. This typically comes from surplus renewable energy production at this University. The highest energy sales occur in April, while the lowest one occur in August (late summer season). The Net Energy Purchased (kWh) represents the net amount of electricity bought after accounting for the energy sold back to the national grid. Peak load (kW) indicates the maximum electrical power demand. Finally, the energy charge in US Dollars for these grid

utility lines is provided in the last column of Table 5.11. Below are two diagrams (Figure 5.13 and Figure 5.14) illustrating the utility grid electricity supplied to the NUST University as simulated by HOMER.

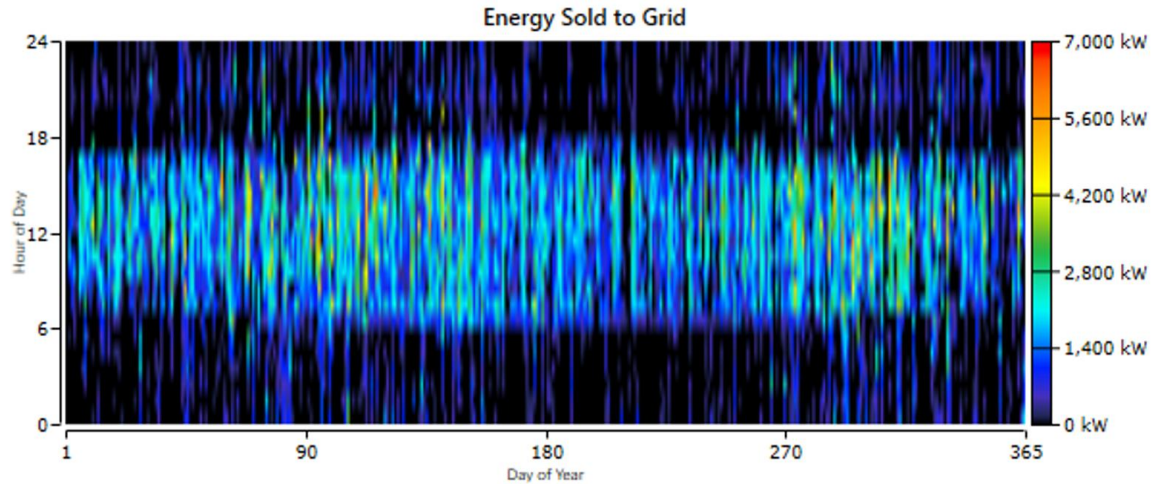


Figure 5.14: Energy Sold to Utility Grid (Renewable Energy)

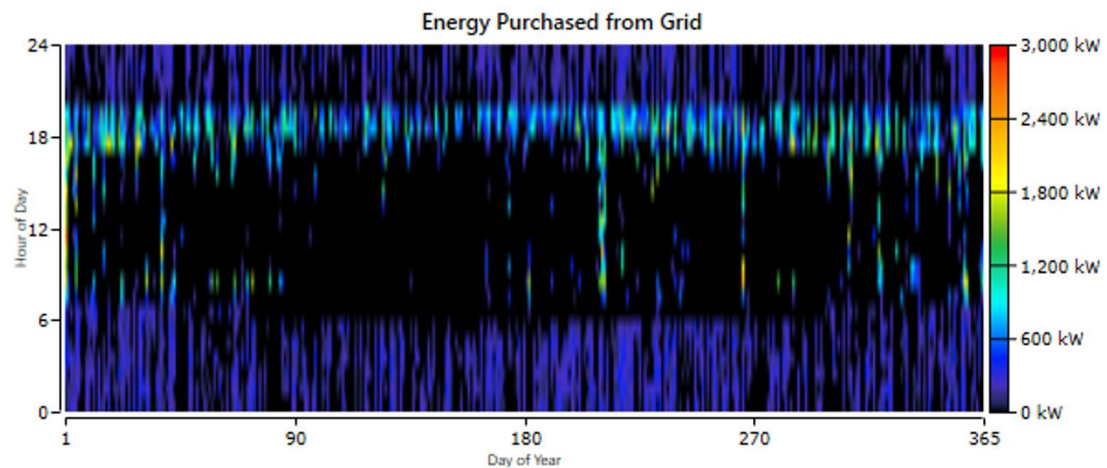


Figure 5.13: Purchased from Utility Grid

5.7 Final Results and Cost for Optimization

The final results for this simulation (HOMER Pro) regarding the costs for the University community in Islamabad incorporate mutually conventional and renewable energy power outputs. It is significant that the calculations may differ from other reports

because of insertion of numerous parameters in this energy optimization method. As a result, the consequences are presented alongside optimization cases. These optimization cases involve further in this simulation of renewable energy at this location. Details of optimization results can be found Appendix 3 respectively. The simulation results for this hybrid microgrid, containing the solar PV and Wind power only are display with the cost of renewable energy electricity. A few of the costs are mentioned in Table 5.12 (below) as well.

Net Present Cost (NPC) represents the total value of all expenses related to a renewable energy system over its lifetime, including capital costs, operations, maintenance, and fuel costs, adjusted to present values using an appropriate discount rate. The discount rate is applied to reduce future cash flows to their present value. For stakeholders who are investing in this University, Islamabad can occur the NPC amounts to US\$11,943,226.36 ([Forbes Currency Converter](#) – 24th July 2024) annually.

The Levelized Cost of Energy (COE) is the average cost of generating electricity over the lifetime of the renewable energy system, expressed per unit of energy produced (\$/kWh). This metric integrates both capital and operating expenses. For this University community, the COE is US\$0.02. The annual operating cost for the microgrid hybrid system, which supplies electricity to the University’s appliances and other needs using both renewable and conventional methods, is US\$242,011.02 only.

Table 5.12: Simulation Result

Total NPC:	US\$11,943,226.36
Levelized COE:	US\$0.02
Operating Cost:	US\$242,011.02

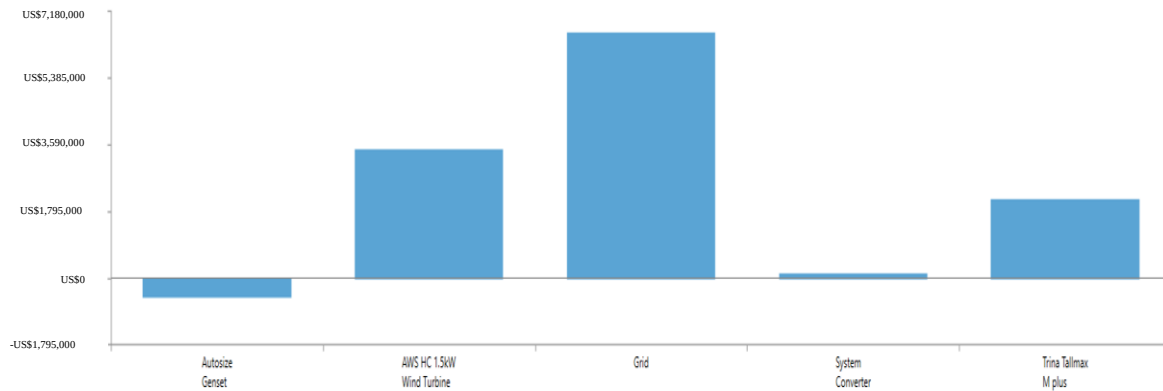


Figure 5.15: University Community Investment Sectors ([Forbes Currency Converter](#) – 24th July 2024)

So, the NPC cost provides a comprehensive view of the long-term financial commitment required for this renewable energy system in microgrid hybrid results, emphasizing the importance of considering future cash flows in present terms. Moreover, the operating cost as mentioned in Table 5.12 (above) demonstrates the ongoing financial cost requirements to maintain the microgrid hybrid system and to ensure continuous electricity supply to this University.

Based on the simulation results from the given simulation (Figure 5.15 – above), the energy costs acquired by the University indicate that the generator's energy production comes with significantly higher financial implications compared to the renewable energy sourced from solar PVs and wind power. Despite the continuous energy generation estimated from these renewable sources, conventional energy remains the primary source of electricity at present. However, the national grid electricity supplied to the NUST University, aside from renewable sources, is mainly derived from utility grid electricity in Pakistan. Figure 5.12 (above) – Neelum-Jhelum Hydroelectricity in Azad Kashmir, Pakistan is an example, which contributes hydroelectric power to the grid. Therefore, transitioning to renewable energy sources could substantially reduce consumption and costs associated with conventional fuel electricity production, while also lessening greenhouse gas emissions. Hence Table 5.13 (below) shows the conventional and renewable energies costs for the University.

According to Table 5.13 in the University community of Islamabad, the Genset generator powered by conventional fossil fuels has a capital cost of US\$1,059,768.00. The annual fuel cost for this generator is US\$0.00 where there is a significant advantage for this community to produce electricity either renewable energy components or utility grid power. Additionally, the salvage cost, or the estimated value of the generator at the end of its useful life, is a negative value. Consequently, fossil fuel generators provide negative value at the end of their operational life and pose a substantial environmental better standard due to the less emission of toxic gases.

Table 5.13: Energy Costs ([Forbes Currency Converter](#) – 24th July 2024)

Component	Capital (US\$)	Replacement (US\$)	O & M (US\$)	Fuel (US\$)	Salvage (US\$)	Total (US\$)
Genset (Conventional Energy)	1,059,768.00	0.00	0.00	0.00	-1,580,479.00	-520,711.00
AWS HC 1.5kW Wind Turbine (Renewable Energy)	747,007.20	944,460.86	2,600,521.04	0.00	-799,827.15	3,492,161.96
Utility Grid	0.00	0.00	6,636,853.75	0.00	0.00	6,636,853.75
System Converter	110,885.80	115,292.09	0.00	0.00	-48,998.19	177,179.69
Trina Tallmax M plus (Renewable Energy)	1,600,549.66	0.00	557,191.83	0.00	0.00	2,157,741.49
Total System	3,518,210.66	1,059,752.95	9,794,566.63	0.00	-2,429,304.34	11,943,225.89

In contrast, the wind power system in the University community (Table 5.13) has a capital cost of US\$747,007.20. The replacement cost is US\$944,460.86, and the operational and maintenance cost is US\$2,600,521.04. Since wind power uses renewable energy, there are no fuel costs. The salvage value at the end of the wind power system's life is negative US\$799,827.15, which offers a financial risk to investors. The total cost for the wind turbine is US\$3,492,161.96.

Similarly, the solar photovoltaic (PV) system, specifically the Trina Tallmax PV model, has a capital cost of US\$1,600,549.66 and operational and maintenance costs of US\$557,191.83 over its 20-year lifespan. The replacement cost is zero, as the system does

not require replacement within this period. However, this also means there is no salvage value, which may concern stakeholders. Despite this, the solar PV system produces no carbon emissions, making it an environmentally friendly alternative to conventional energy.

So, the wind power, storage devices, converter, and solar PVs can sustain zero costs, unlike the generator, within this University community. Consequently, investors and stakeholders face lower risks when funding such energy initiatives because the community consistently relies on renewable energy sources. In contrast, conventional energy requires significant funding, which may deter potential investors. However, if the community highlights renewable energy over fossil fuels, investing in energy production becomes a more attractive prospect. Therefore, the University's energy initiatives present an ideal opportunity for comfortable energy payments compared to other communities in managing microgrid hybrid system.

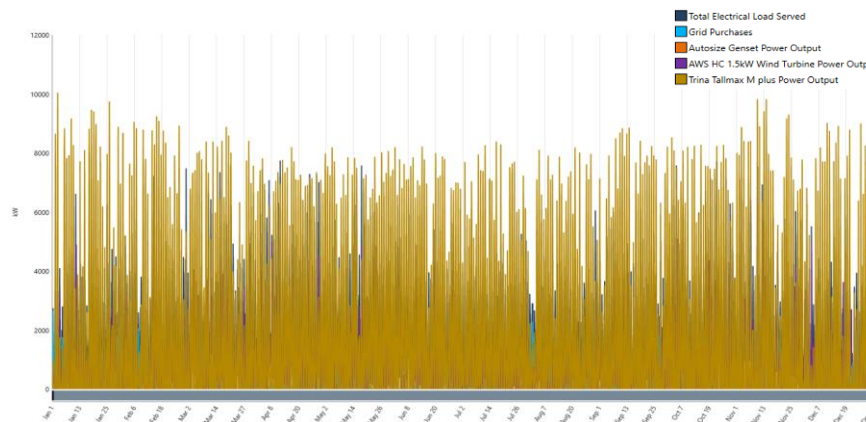


Figure 5.16: University Community Power Output

Figure 5.16 (above) illustrates the annual energy power output in the NUST University. The brown color representing the Trina Tallmark M Plus Power Output (Solar PV) shows its highest power output throughout the year. Meanwhile, the purple color (AWS HC 1.5kW Wind Turbine Power output) has consistently contribute significantly throughout the year with Solar energy. Solar Energy outputs have high peaks during the winter season (October to March), whereas low levels reach from May to September (Summer season).

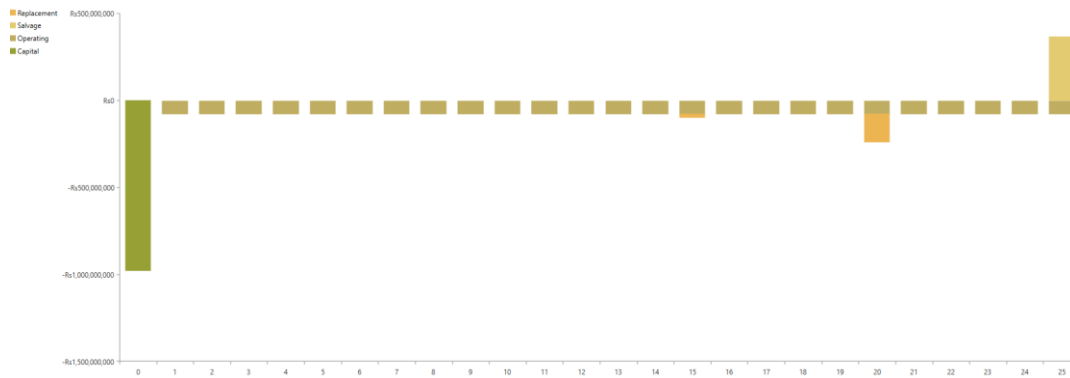


Figure 5.17: Cash Flow Renewable Microgrid

Figure 5.17 – above depicts the cash flow chart displaying replacement, salvage, operating, and capital costs of the hybrid microgrid renewable plant over a 25-year period. This chart illustrates the financial aspects of the renewable energy system implemented in this University. Each cost category is represented by a distinct color, facilitating easy identification and interpretation. All costs, except salvage, are shown as downward bars, indicating cash outflow for expenses. On the contrary, the upward bars representing salvage cost, indicate in income generated from the selling methods.

- **Replacement:** Negative cash flow in year zero (orange bar). This indicates an initial investment or replacement cost. The initial replacement cost (year zero) is significant (negative cash flow).
- **Salvage:** Occasional positive cash flow (beige color bars). Indicates salvage value or end-of-life costs. Salvage costs occur sporadically.
- **Operating:** Negative cash flows (brown color bars). Represents ongoing operational costs. Operating costs fluctuate.
- **Capital Fundings:** Negative cash flows (green color bars). Indicates capital funding or investment received. Capital funding contributes by stakeholders.

The cash flow for the initial year (Year 0) includes the capital investment for NUST University's renewable and conventional energy systems. Stakeholders according to Figure 5.17 (above) are expected to invest US\$3,518,210.65 (PKR980,002,968.97–Investing

amount given by simulation) initially ([Forbes Currency Converter](#) – 24th July 2024). This capital value is for NUST University components mostly in renewable energy electricity production over 25 years. Hence, integrating renewable energy can help to reduce the number of tons of carbon footprint.

For the first 15 years, the replacement costs occur in these energy components' costing. In fact, the replacements cost of these components in the year 15 (US\$80,084.18) and 20 (US\$581,005.60) is ([Forbes Currency Converter](#) – 24th July 2024) US\$661,089.78 (PKR184,170,298.61). At the end of the 25-year period, the energy investment's salvage value will yield a profit of US\$1,323,510.43 (PKR368,665,857.64) for the stakeholders' discount payback.

Figure 5.18 (below) illustrates the electric production of renewable power output, showing megawatt hours (MWh) against monthly durations. According to this diagram, solar PVs generate the highest electricity output compared to the utility grid and wind power at the university. Wind power turbines also contribute to the renewable energy supply, although to a lesser extent, which may present challenges. The utility grid produces less electricity at this university, although it remains a sufficient method to generate power without impacting the competition with conventional energy. Nonetheless, all three power sources play a crucial role in achieving higher efficiency in renewable energy production, thus reducing the dependence on conventional energy at this location. Additional parameters according to Figure 5.18 are shown in Table 5.14 below.

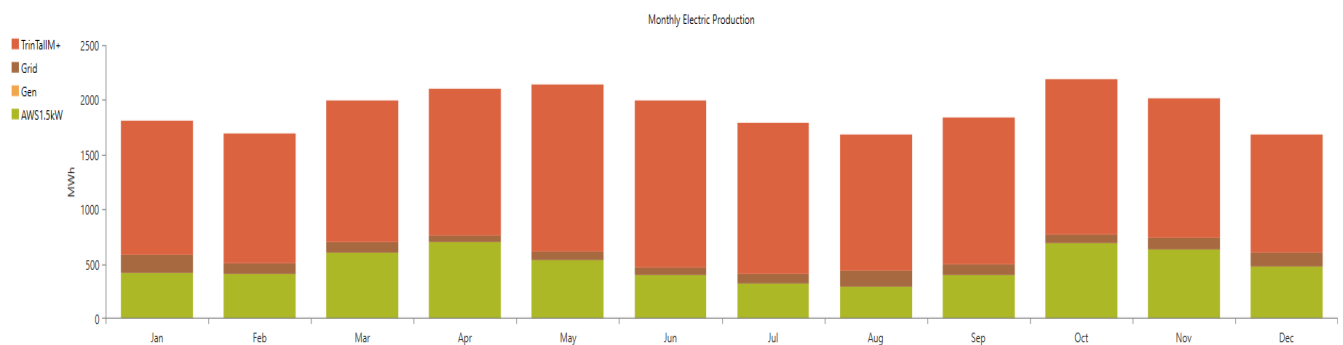


Figure 5.18: Monthly Electric Production (NUST University)

Table 5.14: Energy Output Electric Production Data

Function	kWh/year	%
Production		
Genset Generator	0.00	0.00
AWS HC 1.5kW Wind Turbine	5,886,017.00	25.70
Trina Tallmax M Plus	15,850,440.00	69.20
Utility Grid	1,180,102	5.15
Total Production	22,916,560	100.00
Consumption		
AC Primary Load	8,856,542	54.00
DC Primary Load	0.00	0.00
Utility Grid Sale	7,530,919	46.00
Total Consumption	4105	100.00
Quantity		
Excess Electricity	6,038,502	26.30
Unmet Electric Load	0.00	0.00
Capacity Shortage	0.00	0.00
Quantity	Value	
Renewable Fraction (%)	92.80	
Maximum Renewable Penetration (%)	318.00	

Stakeholders can consider investing in Solar PVs, infrastructure, and maintenance because it produces more electricity throughout the year compared to wind power and utility grid outputs (kWh/year). Despite the challenges, stakeholders can also promote the efficiency of solar PVs and wind power way higher than conventional energy production. For the Genset generator (not mentioned in Table 5.14) that runs on fossil fuels, stakeholders can optimize its usage by balancing petroleum output and renewable energy sources in the University. The microgrid hybrid system, considering insights from Figure 5.18 (above), can ensure efficient use of all power sources in this area.

The percentages in Table 5.14 (above) show the distribution of energy usage in the University community. The Genset generator according to given simulation contributes 0.00% to the electricity supply, significantly reducing the greenhouse emission. Wind power turbines contribute 25.70%, while solar PV add 69.20%. Consequently, renewable energy sources (wind and solar) account for 94.90% ($69.20 + 25.70$) of the total energy production, highlighting a shift towards sustainable energy and away from harmful emissions associated with conventional energy.

These renewable energy productions primarily serve the DC load due to the microgrid system's AC to DC (rectification) conversion principle. The University community also has an excess electricity of 26.30%, generated from wind power, solar PVs, and the utility grid. This indicates efficient utilization of produced energy. As a result, the renewable fraction and maximum renewable penetration show positive values of 92.80%, emphasizing the effectiveness of the microgrid hybrid system in reducing the greenhouse gasses and hence promoting renewable energy in this University. Finally, addressing emissions of carbon dioxide and other harmful gases poses a challenge in optimizing these hybrid microgrid evaluations. However, these toxic gases can be effectively controlled by utilizing the output electricity from renewable energy sources most of the time. Table 5.15 (below) outlines the gas emission behaviors and control values.

Table 5.15: Gasses Emission Data

Quantity	Values	Units
Carbon Dioxide	745,825.00	kg/year
Carbon Monoxide	0.00	kg/year
Unburned Hydrocarbons	0.00	kg/year
Particulate Matter	0.00	kg/year
Sulfur Dioxide	3,233.00	kg/year
Nitrogen Oxides	1,581.00	kg/year

Carbon dioxide emissions (745,825 kg per Year) are the most significant increment among various gases released during energy production, contributing enormously to climate change. This can lead to contrary conditions for the University's environment. Reducing carbon dioxide emissions is possible through these renewable energy devices. Additionally, expanding green spaces can help mitigate the effects of emissions if conventional energy sources are still in use. Therefore, balancing renewable and conventional energy sources is essential to minimize the carbon footprint and enhance the environment with sustainable practices, ensuring a better future with improved greenery and ecological health.

Implementing a microgrid hybrid system can significantly improve environmental conditions by integrating renewable energy sources alongside conventional energy. This approach not only reduces carbon dioxide emissions, which are the primary contributors to

climate change, but also promotes the growth of green spaces. The hybrid system ensures a balanced energy production, leveraging the strengths of both renewable and conventional energy to minimize the carbon footprint. By implementing such systems, communities can achieve sustainable energy solutions, enhancing the present environment and safeguarding it for future generations.

CHAPTER 6: SUMMARY OF RESEARCH WORK

The assessment of the microgrid hybrid device with techno-economic implementation has generated significant perceptions through the utilization of the HOMER software tool, offering insights into results and feasibility conditions. In this evaluation, the hybrid microgrid equipment has primarily focused on the efficiency of solar PV and wind power in charging electricity within the University. However, despite the implementation of these renewable energy sources, the electricity usage in the University has not met the desired standard, demanding continued reliance on fossil fuels and their associated emissions of toxic gases.

Despite the ongoing dependence on fossil fuels, there is potential for resolving this issue in the future by increasing the production of renewable energy during specific periods throughout the year. Both solar PV and wind power units have specific times during which they generate electricity from excess renewable energy units. These units can also be stored using storage devices within the small region. The HOMER software enables simulation of these conditions and prediction of their usage in advance, aligning with the objectives outlined in the research thesis.

CHAPTER 7: CONCLUSION

The feasibility analysis of this hybrid microgrid device both contains with Solar PV and Wind Power as its renewable energy implemented in the NUST University, Islamabad serves as evidence to the community's commitment to be verify as sustainable practices. By integrating these renewable energy sources into the University community's energy infrastructure, they are taking proactive steps towards reducing greenhouse gasses emissions and later fostering a cleaner and neat environment. However, it is vital to strike a balance between renewable energy and conventional energy in this community, derived by primarily energy from fossil fuels or else petroleum segments. Proactive renewable energy only can stance much more challenges by ignoring conventional energy, particularly in terms of maintaining uninterrupted energy supply.

The evaluations conducted using the HOMER software by Computer-Aided machine system have provided valuable perceptions into optimizing the utilization of wind power and solar photovoltaic (PV) cells, thereby collected energy and store into the storage devices afterwards. These insights are essential for forecasting the performance of renewable energy sources and ensuring the reliability and stability of this microgrid hybrid system. While obtaining precise data for renewable energy output predictions which can be or sometimes be challenging, these estimations provided by HOMER software can enable effective planning and management of this microgrid in this University community.

Renewable energy sources like solar PV and wind power offer immense potential benefits for clean energy generation. Solar PV technology is a good way for energy from the sunlight with a peak energy conversion typically occurring during daylight hours. In contrast, wind power can be also utilizing the kinetic energy of the wind movement by flowing of the air volume and to generate electricity afterwards by using its turbine objective performance. This wind power is a viable option for regions with consistent wind patterns. By integrating these renewable energy sources into the microgrid, this University community of Islamabad can significantly reduce its carbon footprint and dependence on fossil fuels later if it is necessary.

The storage cells sector plays a crucial role too in ensuring the stability and reliability system of the hybrid microgrid. These storage devices, typically batteries and inverters can enable the efficient storage and distribution of energy generated by renewable sources. During periods of low renewable energy generation or a high demand of renewable energy's factors, stored energy can be utilized to supplement the grid and to ensure the uninterrupted power supply to the University community.

In terms of feasibility conditions, the implementation of a hybrid microgrid requires consideration of various factors, including technical feasibility, economic capability, and the environmental impact for the renewable energy sources. The HOMER software has been an instrumental tool in assessing these feasibility conditions, providing valuable insights into the cost-effectiveness and sustainability of this microgrid project.

From an economic standpoint, the cash flow analysis presented in Figure 26 highlights the financial implications of the hybrid microgrid project over a 25-year period. The downward bars representing replacement, operating, and capital costs signify cash outflows, while the upward bars indicating salvage represent income generated from selling. This analysis underscores the long-term financial commitment required for the successful implementation and operation of the microgrid solutions.

The hybrid microgrid project in the NUST University, Islamabad represents a significant step towards sustainable energy development. By influence of these renewable energy sources and optimizing their utilization and reducing the conventional energy to that percentage, this University can reduce the emissions, enhance energy resilience, and contribute to a cleaner and greener future. These perceptions gained from these feasibility analyses and hence become as an economic evaluation. This statement can provide valuable guidance for the future renewable energy projects, both within the University and beyond globally.

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APPENDIX

Appendix 1: Typical Electric Load Weekday Data

Hour	January	February	March	April	May	June	July	August	September	October	November	December
0	2400.00	2400.00	2400.00	2400.00	2400.00	2400.00	2400.00	2400.00	2400.00	2400.00	2400.00	2400.00
1	2400.00	2400.00	2400.00	2400.00	2400.00	2400.00	2400.00	2400.00	2400.00	2400.00	2400.00	2400.00
2	2400.00	2400.00	2400.00	2400.00	2400.00	2400.00	2400.00	2400.00	2400.00	2400.00	2400.00	2400.00
3	2400.00	2400.00	2400.00	2400.00	2400.00	2400.00	2400.00	2400.00	2400.00	2400.00	2400.00	2400.00
4	2400.00	2400.00	2400.00	2400.00	2400.00	2400.00	2400.00	2400.00	2400.00	2400.00	2400.00	2400.00
5	2400.00	2400.00	2400.00	2400.00	2400.00	2400.00	2400.00	2400.00	2400.00	2400.00	2400.00	2400.00
6	2400.00	2400.00	2400.00	2400.00	2400.00	2400.00	2400.00	2400.00	2400.00	2400.00	2400.00	2400.00
7	8000.00	8000.00	8000.00	8000.00	8000.00	8000.00	8000.00	8000.00	8000.00	8000.00	8000.00	8000.00
8	16000.00	16000.00	16000.00	16000.00	16000.00	16000.00	16000.00	16000.00	16000.00	16000.00	16000.00	16000.00
9	16000.00	16000.00	16000.00	16000.00	16000.00	16000.00	16000.00	16000.00	16000.00	16000.00	16000.00	16000.00
10	16000.00	16000.00	16000.00	16000.00	16000.00	16000.00	16000.00	16000.00	16000.00	16000.00	16000.00	16000.00
11	16000.00	16000.00	16000.00	16000.00	16000.00	16000.00	16000.00	16000.00	16000.00	16000.00	16000.00	16000.00
12	16000.00	16000.00	16000.00	16000.00	16000.00	16000.00	16000.00	16000.00	16000.00	16000.00	16000.00	16000.00
13	16000.00	16000.00	16000.00	16000.00	16000.00	16000.00	16000.00	16000.00	16000.00	16000.00	16000.00	16000.00
14	16000.00	16000.00	16000.00	16000.00	16000.00	16000.00	16000.00	16000.00	16000.00	16000.00	16000.00	16000.00
15	16000.00	16000.00	16000.00	16000.00	16000.00	16000.00	16000.00	16000.00	16000.00	16000.00	16000.00	16000.00
16	16000.00	16000.00	16000.00	16000.00	16000.00	16000.00	16000.00	16000.00	16000.00	16000.00	16000.00	16000.00
17	12800.00	12800.00	12800.00	12800.00	12800.00	12800.00	12800.00	12800.00	12800.00	12800.00	12800.00	12800.00
18	10400.00	10400.00	10400.00	10400.00	10400.00	10400.00	10400.00	10400.00	10400.00	10400.00	10400.00	10400.00
19	8000.00	8000.00	8000.00	8000.00	8000.00	8000.00	8000.00	8000.00	8000.00	8000.00	8000.00	8000.00
20	2400.00	2400.00	2400.00	2400.00	2400.00	2400.00	2400.00	2400.00	2400.00	2400.00	2400.00	2400.00
21	2400.00	2400.00	2400.00	2400.00	2400.00	2400.00	2400.00	2400.00	2400.00	2400.00	2400.00	2400.00
22	2400.00	2400.00	2400.00	2400.00	2400.00	2400.00	2400.00	2400.00	2400.00	2400.00	2400.00	2400.00
23	2400.00	2400.00	2400.00	2400.00	2400.00	2400.00	2400.00	2400.00	2400.00	2400.00	2400.00	2400.00

Appendix 2: Typical Electric Load Weekend Data

Hour	January	February	March	April	May	June	July	August	September	October	November	December
0	2400.00	2400.00	2400.00	2400.00	2400.00	2400.00	2400.00	2400.00	2400.00	2400.00	2400.00	2400.00
1	2400.00	2400.00	2400.00	2400.00	2400.00	2400.00	2400.00	2400.00	2400.00	2400.00	2400.00	2400.00
2	2400.00	2400.00	2400.00	2400.00	2400.00	2400.00	2400.00	2400.00	2400.00	2400.00	2400.00	2400.00
3	2400.00	2400.00	2400.00	2400.00	2400.00	2400.00	2400.00	2400.00	2400.00	2400.00	2400.00	2400.00
4	2400.00	2400.00	2400.00	2400.00	2400.00	2400.00	2400.00	2400.00	2400.00	2400.00	2400.00	2400.00
5	2400.00	2400.00	2400.00	2400.00	2400.00	2400.00	2400.00	2400.00	2400.00	2400.00	2400.00	2400.00
6	2400.00	2400.00	2400.00	2400.00	2400.00	2400.00	2400.00	2400.00	2400.00	2400.00	2400.00	2400.00
7	6000.00	6000.00	6000.00	6000.00	6000.00	6000.00	6000.00	6000.00	6000.00	6000.00	6000.00	6000.00
8	8000.00	8000.00	8000.00	8000.00	8000.00	8000.00	8000.00	8000.00	8000.00	8000.00	8000.00	8000.00
9	8000.00	8000.00	8000.00	8000.00	8000.00	8000.00	8000.00	8000.00	8000.00	8000.00	8000.00	8000.00
10	8000.00	8000.00	8000.00	8000.00	8000.00	8000.00	8000.00	8000.00	8000.00	8000.00	8000.00	8000.00
11	8000.00	8000.00	8000.00	8000.00	8000.00	8000.00	8000.00	8000.00	8000.00	8000.00	8000.00	8000.00
12	8000.00	8000.00	8000.00	8000.00	8000.00	8000.00	8000.00	8000.00	8000.00	8000.00	8000.00	8000.00
13	8000.00	8000.00	8000.00	8000.00	8000.00	8000.00	8000.00	8000.00	8000.00	8000.00	8000.00	8000.00
14	8000.00	8000.00	8000.00	8000.00	8000.00	8000.00	8000.00	8000.00	8000.00	8000.00	8000.00	8000.00
15	8000.00	8000.00	8000.00	8000.00	8000.00	8000.00	8000.00	8000.00	8000.00	8000.00	8000.00	8000.00
16	8000.00	8000.00	8000.00	8000.00	8000.00	8000.00	8000.00	8000.00	8000.00	8000.00	8000.00	8000.00
17	6800.00	6800.00	6800.00	6800.00	6800.00	6800.00	6800.00	6800.00	6800.00	6800.00	6800.00	6800.00
18	6400.00	6400.00	6400.00	6400.00	6400.00	6400.00	6400.00	6400.00	6400.00	6400.00	6400.00	6400.00
19	5000.00	5000.00	5000.00	5000.00	5000.00	5000.00	5000.00	5000.00	5000.00	5000.00	5000.00	5000.00
20	2400.00	2400.00	2400.00	2400.00	2400.00	2400.00	2400.00	2400.00	2400.00	2400.00	2400.00	2400.00
21	2400.00	2400.00	2400.00	2400.00	2400.00	2400.00	2400.00	2400.00	2400.00	2400.00	2400.00	2400.00
22	2400.00	2400.00	2400.00	2400.00	2400.00	2400.00	2400.00	2400.00	2400.00	2400.00	2400.00	2400.00
23	2400.00	2400.00	2400.00	2400.00	2400.00	2400.00	2400.00	2400.00	2400.00	2400.00	2400.00	2400.00

Appendix 3: Optimization Costs [\(Forbes Currency Converter – 24th July 2024\)](#)

Architecture						Cost				System		Generator						Solar PV TrinTallM+		Wind Power Turbine AWS1.5kW			Utility Grid		1kWh Lithium		
TrinTallM+ (kW)	AWS1.5kW	Grid (kW)	Gen (kW)	1kWh Lithium	Converter (kW)	NPC (US\$)	COE (US\$)	Operating cost (US\$/yr)	Initial capital (US\$)	Ren Frac (%)	Total Fuel (L/yr)	Hours	Production (kWh)	Fuel (L)	O&M Cost (US\$/yr)	Fuel Cost (US\$/yr)	Capital Cost (US\$)	Production (kWh/yr)	O&M Cost (US\$)	Energy Purchased (kWh)	Emerg Sold (kWh)	Autonomy (Hours)	Nominal Capacity (kWh)	Usable Nominal Capacity (kWh)			
9174.00	2312.00	20000.00	4100.00		2860.00	11,954,700.00	0.02	241,966.00	3,518,200.00	92.80	0.00	0.00	0.00	0.00	0.00	0.00	1,600,549.63	1585044.000	747,007.20	5886017.00	74,700.72	1180102.00	7530918.00				
9274.00	2133.00	20000.00	4100.00	14	2984.00	11,954,700.00	0.02	243,402.00	3,485,890.00	92.50	0.00	0.00	0.00	0.00	0.00	0.00	1,618,029.89	16023549.00	689,172.30	5430309.00	68,917.23	1224367.00	7437306.00	0.0111	14.00	11.20	
9174.00	2312.00	20000.00			2860.00	12,457,300.00	0.02	287,559.00	2,459,150.00	92.80	0.00						1,600,549.63	15850440.00	747,007.20	5886017.00	74,700.72	1180102.00	7530918.00				
9185.00	2283.00	20000.00		1	3009.00	12,457,300.00	0.02	287,559.00	2,455,560.00	92.90	0.00						1,602,468.01	15869438.00	737,637.30	5812188.00	73,763.73	1185934.00	7810836.00	1.000791	1.00	0.80	
13993.00		20000.00	4100.00		3401.00	16,765,300.00	0.03	376,950.00	3,625,900.00	82.80	0.00	0.00	0.00	0.00	0.00	0.00	2,441,437.80	24177860.00				2453216.00	5378226.00				
13905.00		20000.00	4100.00	4	3345.00	16,765,300.00	0.03	376,950.00	3,625,900.00	82.50	0.00	0.00	0.00	0.00	0.00	0.00	2,425,984.60	24024824.00				2457493.00	5214856.00	0.00317	4.00	3.20	
13993.00		20000.00			3401.00	17,303,800.00	0.03	423,620.00	2,574,030.00	82.80	0.00						2,441,437.80	24177860.00				2453216.00	5378226.00				
13905.00		20000.00		4	3345.00	17,303,800.00	0.04	423,620.00	2,556,080.00	82.50	0.00						2,425,984.60	24024824.00				2457493.00	5214856.00	0.00317	4.00	3.20	
	6066.00	20000.00	4100.00			23,873,500.00	0.04	599,530.00	3,019,190.00	85.10	0.00	0.00	0.00	0.00	0.00	0.00			1,959,924.49	15443157.00	195,992.46	2707599.00	9294214.00				
	5423.00	20000.00	4100.00	62	18.00	23,981,200.00	0.04	606,710.00	2,825,330.00	82.60	0.00	0.00	0.00	0.00	0.00	0.00			1,752,171.36	13806173.00	175,217.13	2898430.00	7848060.00	0.0491	62.00	49.60	
	6066.00	20000.00				24,376,100.00	0.04	642,610.00	1,960,140.00	85.10	0.00								1,959,924.49	15443157.00	195,992.46	2707599.00	9294214.00				
	5423.00	20000.00		62	18.00	24,519,700.00	0.04	653,380.00	1,766,280.00	82.60	0.00								1,752,171.36	13806173.00	175,217.13	2898430.00	7848060.00	0.0491	62.00	49.60	
		20000.00	4100.00			49,183,000.00	0.16	1,385,740.00	1,059,050.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00						8856543.00	0.00				
		20000.00	4100.00	158	28.60	49,183,000.00	0.16	1,385,740.00	1,091,360.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00						8856543.00	0.00	0.125	158.00	126.00	
		20000.00				49,901,000.00	0.16	1,432,410.00	0.00	0.00	0.00											8856543.00	0.00				
		20000.00		158	28.60	49,901,000.00	0.16	1,432,410.00	32,310.00	0.00	0.00											8856543.00	0.00	0.125	158.00	126.00	

NPC – Net Present Cost (US\$)

COE – Cost of Energy

Ren Frac (%) – Renewable Fraction (%)

O & M Cost – Operational and Maintenance Cost

Appendix 4: Scaled Data Daily Electric Load



LIST OF PUBLICATIONS

Heliyon, HELIYON-D-24-24781(R1), Evaluation of Tecno-Economic Design and Implementation of Solar-wind hybrid Microgrid for a small community, Tracking System:
https://www.editorialmanager.com/heliyon/default.aspx?_hstc=25856994.e3cf87a874036f2441637aafd2b13628.1713788169089.1716993021292.1717139286029.7&_hssc=25856994.2.1717139286029&_hsfp=2485312484 Status: 1stRevision.